

11 QUANTUM DYNAMICS

So far, practically everything we have done belongs to the subject that might properly be called **quantum statics**, in which the *potential energy function is independent of time*: $V(\mathbf{r}, t) = V(\mathbf{r})$. In that case the (time-dependent) Schrödinger equation,

$$i\hbar \frac{\partial \Psi}{\partial t} = \hat{H}\Psi, \quad (11.1)$$

can be solved by separation of variables:

$$\Psi(\mathbf{r}, t) = \psi(\mathbf{r})e^{-iEt/\hbar}, \quad (11.2)$$

where $\psi(\mathbf{r})$ satisfies the time-*independent* Schrödinger equation,

$$\hat{H}\psi = E\psi. \quad (11.3)$$

Because the time dependence of separable solutions is carried by the exponential factor ($e^{-iEt/\hbar}$), which cancels out when we construct the physically relevant quantity $|\Psi|^2$, all probabilities and expectation values (for such states) are constant in time. By forming *linear combinations* of these stationary states we obtain wave functions with more interesting time dependence,

$$\Psi(\mathbf{r}, t) = \sum c_n \psi_n(\mathbf{r}) e^{-iE_n t/\hbar}, \quad (11.4)$$

but even then the possible values of the energy (E_n), and their respective probabilities ($|c_n|^2$), are constant.

If we want to allow for **transitions** (**quantum jumps**, as they are sometimes called) between one energy level and another, we must introduce a *time-dependent* potential (**quantum dynamics**). There are precious few exactly solvable problems in quantum dynamics. However, if the time-dependent part of the Hamiltonian is *small* (compared to the time-independent part), it can be treated as a perturbation. The main purpose of this chapter is to develop **time-dependent perturbation theory**, and study its most important application: the emission or absorption of radiation by an atom.

Problem 11.1 Why isn't it *trivial* to solve the time-dependent Schrödinger equation (11.1), in its dependence on t ? After all, it's a first-order differential equation.

(a) How would you solve the equation

$$\frac{df}{dt} = k f$$

(for $f(t)$), if k were a constant?

- (b) What if k is itself a function of t ? (Here $k(t)$ and $f(t)$ might also depend on other variables, such as $\mathbf{r}$ —it doesn't matter.)
- (c) Why not do the same thing for the Schrödinger equation (with a time-dependent Hamiltonian)? To see that this doesn't work, consider the simple case

$$\hat{H}(t) = \begin{cases} \hat{H}_1, & (0 < t < \tau), \\ \hat{H}_2, & (t > \tau), \end{cases}$$

where $\hat{H}_1$ and $\hat{H}_2$ are themselves time-independent. If the solution in part (b) held for the Schrödinger equation, the wave function at time $t > \tau$ would be

$$\Psi(t) = e^{-i[\hat{H}_1\tau + \hat{H}_2(t-\tau)]/\hbar} \Psi(0),$$

but of course we could also write

$$\Psi(t) = e^{-i\hat{H}_2(t-\tau)/\hbar} \Psi(\tau) = e^{-i\hat{H}_2(t-\tau)/\hbar} e^{-i\hat{H}_1\tau/\hbar} \Psi(0).$$

Why are these generally *not* the same? [This is a subtle matter; if you want to pursue it further, see Problem 11.23.]

11.1 TWO-LEVEL SYSTEMS

To begin with, let us suppose that there are just *two* states of the (unperturbed) system, ψ_a and ψ_b . They are eigenstates of the unperturbed Hamiltonian, $\hat{H}^0$:

$$\hat{H}^0\psi_a = E_a\psi_a, \quad \text{and} \quad \hat{H}^0\psi_b = E_b\psi_b, \quad (11.5)$$

and they are orthonormal:

$$\langle \psi_i | \psi_j \rangle = \delta_{ij}, \quad (i, j = a, b). \quad (11.6)$$

Any state can be expressed as a linear combination of them; in particular,

$$\Psi(0) = c_a\psi_a + c_b\psi_b. \quad (11.7)$$

The states ψ_a and ψ_b might be position-space wave functions, or spinors, or something more exotic—it doesn't matter. It is the *time* dependence that concerns us here, so when I write $\Psi(t)$, I simply mean the state of the system at time t . In the absence of any perturbation, each component evolves with its characteristic wiggle factor:

$$\Psi(t) = c_a\psi_a e^{-iE_a t/\hbar} + c_b\psi_b e^{-iE_b t/\hbar}. \quad (11.8)$$

Informally, we say that $|c_a|^2$ is the “probability that the particle is in state ψ_a ”—by which we *really* mean the probability that a measurement of the energy would yield the value E_a . Normalization of Ψ requires, of course, that

$$|c_a|^2 + |c_b|^2 = 1. \quad (11.9)$$

11.1.1 The Perturbed System

Now suppose we turn on a time-dependent perturbation, $\hat{H}'(t)$. Since ψ_a and ψ_b constitute a complete set, the wave function $\Psi(t)$ can still be expressed as a linear combination of them. The only difference is that c_a and c_b are now *functions of t* :

$$\Psi(t) = c_a(t)\psi_a e^{-iE_a t/\hbar} + c_b(t)\psi_b e^{-iE_b t/\hbar}. \quad (11.10)$$

(I could absorb the exponential factors into $c_a(t)$ and $c_b(t)$, and some people prefer to do it this way, but I think it is nicer to keep visible the part of the time dependence that would be present even *without* the perturbation.) The whole problem is to determine c_a and c_b , as functions of time. If, for example, the particle started out in the state ψ_a ($c_a(0) = 1$, $c_b(0) = 0$), and at some later time t_1 we find that $c_a(t_1) = 0$, $c_b(t_1) = 1$, we shall report that the system underwent a transition from ψ_a to ψ_b .

We solve for $c_a(t)$ and $c_b(t)$ by demanding that $\Psi(t)$ satisfy the time-dependent Schrödinger equation,

$$\hat{H}\Psi = i\hbar \frac{\partial \Psi}{\partial t}, \quad \text{where} \quad \hat{H} = \hat{H}^0 + \hat{H}'(t). \quad (11.11)$$

From Equations 11.10 and 11.11, we find:

$$\begin{aligned} c_a \left(\hat{H}^0 \psi_a \right) e^{-iE_a t/\hbar} + c_b \left(\hat{H}^0 \psi_b \right) e^{-iE_b t/\hbar} + c_a \left(\hat{H}' \psi_a \right) e^{-iE_a t/\hbar} \\ + c_b \left(\hat{H}' \psi_b \right) e^{-iE_b t/\hbar} = i\hbar \left[\dot{c}_a \psi_a e^{-iE_a t/\hbar} + \dot{c}_b \psi_b e^{-iE_b t/\hbar} \right. \\ \left. + c_a \psi_a \left(-\frac{iE_a}{\hbar} \right) e^{-iE_a t/\hbar} + c_b \psi_b \left(-\frac{iE_b}{\hbar} \right) e^{-iE_b t/\hbar} \right]. \end{aligned}$$

In view of Equation 11.5, the first two terms on the left cancel the last two terms on the right, and hence

$$c_a \left(\hat{H}' \psi_a \right) e^{-iE_a t/\hbar} + c_b \left(\hat{H}' \psi_b \right) e^{-iE_b t/\hbar} = i\hbar \left(\dot{c}_a \psi_a e^{-iE_a t/\hbar} + \dot{c}_b \psi_b e^{-iE_b t/\hbar} \right). \quad (11.12)$$

To isolate $\dot{c}_a$, we use the standard trick: Take the inner product with ψ_a , and exploit the orthogonality of ψ_a and ψ_b (Equation 11.6):

$$c_a \left\langle \psi_a \left| \hat{H}' \right| \psi_a \right\rangle e^{-iE_a t/\hbar} + c_b \left\langle \psi_a \left| \hat{H}' \right| \psi_b \right\rangle e^{-iE_b t/\hbar} = i\hbar \dot{c}_a e^{-iE_a t/\hbar}.$$

For short, we define

$$H'_{ij} \equiv \left\langle \psi_i \left| \hat{H}' \right| \psi_j \right\rangle; \quad (11.13)$$

note that the hermiticity of $\hat{H}'$ entails $H'_{ji} = (H'_{ij})^*$. Multiplying through by $-(i/\hbar) e^{iE_a t/\hbar}$, we conclude that:

$$\dot{c}_a = -\frac{i}{\hbar} \left[c_a H'_{aa} + c_b H'_{ab} e^{-i(E_b - E_a)t/\hbar} \right]. \quad (11.14)$$

Similarly, the inner product with ψ_b picks out $\dot{c}_b$:

$$c_a \left\langle \psi_b \left| \hat{H}' \right| \psi_a \right\rangle e^{-iE_a t/\hbar} + c_b \left\langle \psi_b \left| \hat{H}' \right| \psi_b \right\rangle e^{-iE_b t/\hbar} = i\hbar \dot{c}_b e^{-iE_b t/\hbar},$$

and hence

$$\dot{c}_b = -\frac{i}{\hbar} \left[c_b H'_{bb} + c_a H'_{ba} e^{i(E_b - E_a)t/\hbar} \right]. \quad (11.15)$$

Equations 11.14 and 11.15 determine $c_a(t)$ and $c_b(t)$; taken together, they are completely equivalent to the (time-dependent) Schrödinger equation, for a two-level system. Typically, the diagonal matrix elements of $\hat{H}'$ vanish (see Problem 11.5 for the general case):

$$H'_{aa} = H'_{bb} = 0. \quad (11.16)$$

If so, the equations simplify:

$$\dot{c}_a = -\frac{i}{\hbar} H'_{ab} e^{-i\omega_0 t} c_b, \quad \dot{c}_b = -\frac{i}{\hbar} H'_{ba} e^{i\omega_0 t} c_a, \quad (11.17)$$

where

$$\omega_0 \equiv \frac{E_b - E_a}{\hbar}. \quad (11.18)$$

(I'll assume that $E_b \geq E_a$, so $\omega_0 \geq 0$.)

- * **Problem 11.2** A hydrogen atom is placed in a (time-dependent) electric field $\mathbf{E} = E(t)\hat{k}$. Calculate all four matrix elements H'_{ij} of the perturbation $\hat{H}' = eEz$ between the ground state ($n = 1$) and the (quadruply degenerate) first excited states ($n = 2$). Also show that $H'_{ii} = 0$ for all five states. *Note:* There is only one integral to be done here, if you exploit oddness with respect to z ; only one of the $n = 2$ states is “accessible” from the ground state by a perturbation of this form, and therefore the system functions as a two-state configuration—assuming transitions to higher excited states can be ignored.

- * **Problem 11.3** Solve Equation 11.17 for the case of a *time-independent* perturbation, assuming that $c_a(0) = 1$ and $c_b(0) = 0$. Check that $|c_a(t)|^2 + |c_b(t)|^2 = 1$. *Comment:* Ostensibly, this system oscillates between “pure ψ_a ” and “some ψ_b .” Doesn't this contradict my general assertion that no transitions occur for time-independent perturbations? No, but the reason is rather subtle: In this case ψ_a and ψ_b are not, and never were, eigenstates of the Hamiltonian—a measurement of the energy *never* yields E_a or E_b . In time-dependent perturbation theory we typically contemplate turning *on* the perturbation for a while, and then turning it *off* again, in order to examine the system. At the beginning, and at the end, ψ_a and ψ_b are eigenstates of the exact Hamiltonian, and only in this context does it make sense to say that the system underwent a transition from one to the other. For the present problem, then, assume that the perturbation was turned on at time $t = 0$, and off again at time T —this doesn't affect the *calculations*, but it allows for a more sensible interpretation of the result.

- ** **Problem 11.4** Suppose the perturbation takes the form of a delta function (in time):

$$\hat{H}' = \hat{U}\delta(t);$$

assume that $U_{aa} = U_{bb} = 0$, and let $U_{ab} = U_{ba}^* \equiv \alpha$. If $c_a(-\infty) = 1$ and $c_b(-\infty) = 0$, find $c_a(t)$ and $c_b(t)$, and check that $|c_a(t)|^2 + |c_b(t)|^2 = 1$. What is the net probability ($P_{a \rightarrow b}$ for $t \rightarrow \infty$) that a transition occurs? *Hint:* You might want to treat the delta function as the limit of a sequence of rectangles. *Answer:* $P_{a \rightarrow b} = \sin^2(|\alpha|/\hbar)$.

11.1.2 Time-Dependent Perturbation Theory

So far, everything is *exact*: We have made no assumption about the *size* of the perturbation. But if $\hat{H}'$ is “small,” we can solve Equation 11.17 by a process of successive approximations, as follows. Suppose the particle starts out in the lower state:

$$c_a(0) = 1, \quad c_b(0) = 0. \quad (11.19)$$

If there were *no perturbation at all*, they would stay this way forever:

Zeroth Order:

$$c_a^{(0)}(t) = 1, \quad c_b^{(0)}(t) = 0. \quad (11.20)$$

(I'll use a superscript in parentheses to indicate the order of the approximation.)

To calculate the first-order approximation, we insert the zeroth-order values on the right side of Equation 11.17:

First Order:

$$\begin{aligned} \frac{dc_a^{(1)}}{dt} &= 0 \Rightarrow c_a^{(1)}(t) = 1; \\ \frac{dc_b^{(1)}}{dt} &= -\frac{i}{\hbar} H'_{ba} e^{i\omega_0 t} \Rightarrow c_b^{(1)} = -\frac{i}{\hbar} \int_0^t H'_{ba}(t') e^{i\omega_0 t'} dt'. \end{aligned} \quad (11.21)$$

Now we insert *these* expressions on the right side of Equation 11.17 to obtain the *second-order* approximation:

Second Order:

$$\begin{aligned} \frac{dc_a^{(2)}}{dt} &= -\frac{i}{\hbar} H'_{ab} e^{-i\omega_0 t} \left(-\frac{i}{\hbar} \right) \int_0^t H'_{ba}(t') e^{i\omega_0 t'} dt' \Rightarrow \\ c_a^{(2)}(t) &= 1 - \frac{1}{\hbar^2} \int_0^t H'_{ab}(t') e^{-i\omega_0 t'} \left[\int_0^{t'} H'_{ba}(t'') e^{i\omega_0 t''} dt'' \right] dt', \end{aligned} \quad (11.22)$$

while c_b is unchanged ($c_b^{(2)}(t) = c_b^{(1)}(t)$). (Notice that $c_a^{(2)}(t)$ *includes* the zeroth-order term; the second-order *correction* would be the integral part alone.)

In principle, we could continue this ritual indefinitely, always inserting the n th-order approximation into the right side of Equation 11.17, and solving for the $(n+1)$ th order. The zeroth order contains *no* factors of $\hat{H}'$, the first-order correction contains *one* factor of $\hat{H}'$, the second-order correction has *two* factors of $\hat{H}'$, and so on.¹ The error in the first-order approximation is evident in the fact that $|c_a^{(1)}(t)|^2 + |c_b^{(1)}(t)|^2 \neq 1$ (the *exact* coefficients must, of course, obey Equation 11.9). However, $|c_a^{(1)}(t)|^2 + |c_b^{(1)}(t)|^2$ is equal to 1 *to first order in $\hat{H}'$* , which is all we can expect from a first-order approximation. And the same goes for the higher orders.

Equation 11.21 can be written in the form

$$c_b^{(1)}(t) e^{-i E_b t / \hbar} = -\frac{i}{\hbar} \int_0^t e^{-i E_b (t-t') / \hbar} H'_{ba}(t') e^{-i E_a t' / \hbar} dt' \quad (11.23)$$

(where I've restored the exponential we factored out in Equation 11.10). This suggests a nice pictorial interpretation: reading from right to left, the system remains in state a from time 0 to time t' (picking up the “wobble factor” $e^{-i E_a t' / \hbar}$), makes a transition from state a to state b at time t' , and then remains in state b until time t (picking up the “wobble factor” $e^{-i E_b (t-t') / \hbar}$). This process is represented in Figure 11.1. (Don't take the picture too literally: there is no

¹ Notice that c_a is modified in every *even* order, and c_b in every *odd* order; this would not be true if the perturbation included diagonal terms, or if the system started out in a linear combination of the two states.

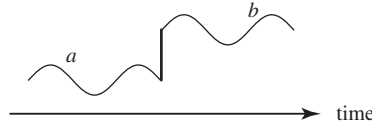


Figure 11.1: Pictorial representation of Equation 11.23.



Figure 11.2: Pictorial representation of Equation 11.24.

sharp transition between these states; in fact, you integrate over all the times t' at which this transition can occur.)

This interpretation of the perturbation series is especially illuminating at higher orders and for multi-level systems, where the expressions become complicated. Consider Equation 11.22, which can be written

$$c_a^{(2)}(t) e^{-i E_a t / \hbar} = e^{-i E_a t / \hbar} + \left(-\frac{i}{\hbar}\right)^2 \int_0^t \int_0^{t'} e^{-i E_a (t-t') / \hbar} \times H'_{ab}(t') e^{-i E_b (t'-t'') / \hbar} H'_{ba}(t'') e^{-i E_a t'' / \hbar} dt'' dt'. \quad (11.24)$$

The two terms here describe a process where the system remains in state a for the entire time, and a second process where the system transitions from a to b at time t'' and then back to a at time t' . Graphically, this is shown in Figure 11.2.

With the insight provided by these pictures, it is easy to write down the general result for a multi-level system:²

$$c_n^{(2)}(t) e^{-i E_n t / \hbar} = \delta_{ni} e^{-i E_i t / \hbar} + \left(\frac{-i}{\hbar}\right) \int_0^t e^{-i E_n (t-t') / \hbar} H_{ni}(t') e^{-i E_i t' / \hbar} dt' + \sum_m \left(\frac{-i}{\hbar}\right)^2 \int_0^t \int_0^{t'} e^{-i E_n (t-t') / \hbar} H_{nm}(t') e^{-i E_m (t'-t'') / \hbar} \times H_{mi}(t'') e^{-i E_i t'' / \hbar} dt'' dt'. \quad (11.25)$$

For $n \neq i$, this is represented by the diagram in Figure 11.3. The first-order term describes a direct transition from i to n , and the second-order term describes a process where the transition occurs via an intermediate (or “virtual”) state m .

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Problem 11.5 Suppose you *don't* assume $H'_{aa} = H'_{bb} = 0$.

- (a) Find $c_a(t)$ and $c_b(t)$ in first-order perturbation theory, for the case $c_a(0) = 1$, $c_b(0) = 0$. Show that $|c_a^{(1)}(t)|^2 + |c_b^{(1)}(t)|^2 = 1$, to first order in $\hat{H}'$.
- (b) There is a nicer way to handle this problem. Let

$$d_a \equiv e^{\frac{i}{\hbar} \int_0^t H'_{aa}(t') dt'} c_a, \quad d_b \equiv e^{\frac{i}{\hbar} \int_0^t H'_{bb}(t') dt'} c_b. \quad (11.26)$$

² Perturbation theory for multi-level systems is treated in Problem 11.24.

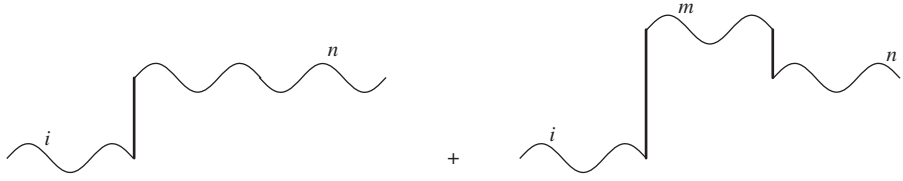


Figure 11.3: Pictorial representation of Equation 11.25 for $n \neq i$.

Show that

$$\dot{d}_a = -\frac{i}{\hbar} e^{i\phi} H'_{ab} e^{-i\omega_0 t} d_b; \quad \dot{d}_b = -\frac{i}{\hbar} e^{-i\phi} H'_{ba} e^{i\omega_0 t} d_a, \quad (11.27)$$

where

$$\phi(t) \equiv \frac{1}{\hbar} \int_0^t [H'_{aa}(t') - H'_{bb}(t')] dt'. \quad (11.28)$$

So the equations for d_a and d_b are identical in structure to Equation 11.17 (with an extra factor $e^{i\phi}$ tacked onto $\hat{H}'$).

(c) Use the method in part (b) to obtain $c_a(t)$ and $c_b(t)$ in first-order perturbation theory, and compare your answer to (a). Comment on any discrepancies.

* **Problem 11.6** Solve Equation 11.17 to second order in perturbation theory, for the general case $c_a(0) = a$, $c_b(0) = b$.

** **Problem 11.7** Calculate $c_a(t)$ and $c_b(t)$, to second order, for the perturbation in Problem 11.3. Compare your answer with the exact result.

* **Problem 11.8** Consider a perturbation to a two-level system with matrix elements

$$H'_{ab} = H'_{ba} = \frac{\alpha}{\sqrt{\pi} \tau} e^{-(t/\tau)^2}, \quad H'_{aa} = H'_{bb} = 0.$$

where τ and α are positive constants with the appropriate units.

(a) According to first-order perturbation theory, if the system starts off in the state $c_a = 1$, $c_b = 0$ at $t = -\infty$, what is the probability that it will be found in the state b at $t = \infty$?

(b) In the limit that $\tau \rightarrow 0$, $H'_{ab} = \alpha \delta(t)$. Compute the $\tau \rightarrow 0$ limit of your expression from part (a) and compare the result of Problem 11.4.

(c) Now consider the opposite extreme: $\omega_0 \tau \gg 1$. What is the limit of your expression from part (a)? *Comment:* This is an example of the adiabatic theorem (Section 11.5.2).

11.1.3 Sinusoidal Perturbations

Suppose the perturbation has sinusoidal time dependence:

$$\hat{H}'(\mathbf{r}, t) = V(\mathbf{r}) \cos(\omega t), \quad (11.29)$$

so that

$$H'_{ab} = V_{ab} \cos(\omega t), \quad (11.30)$$

where

$$V_{ab} \equiv \langle \psi_a | V | \psi_b \rangle. \quad (11.31)$$

(As before, I'll assume the diagonal matrix elements vanish, since this is almost always the case in practice.) To first order (from now on we'll work *exclusively* in first order, and I'll dispense with the superscripts) we have (Equation 11.21):

$$\begin{aligned} c_b(t) &\approx -\frac{i}{\hbar} V_{ba} \int_0^t \cos(\omega t') e^{i\omega_0 t'} dt' = -\frac{i V_{ba}}{2\hbar} \int_0^t \left[e^{i(\omega_0 + \omega)t'} + e^{i(\omega_0 - \omega)t'} \right] dt' \\ &= -\frac{V_{ba}}{2\hbar} \left[\frac{e^{i(\omega_0 + \omega)t} - 1}{\omega_0 + \omega} + \frac{e^{i(\omega_0 - \omega)t} - 1}{\omega_0 - \omega} \right]. \end{aligned} \quad (11.32)$$

That's the *answer*, but it's a little cumbersome to work with. Things simplify substantially if we restrict our attention to driving frequencies (ω) that are very close to the transition frequency (ω_0), so that the second term in the square brackets dominates; specifically, we assume:

$$\omega_0 + \omega \gg |\omega_0 - \omega|. \quad (11.33)$$

This is not much of a limitation, since perturbations at *other* frequencies have a negligible probability of causing a transition anyway. Dropping the first term, we have

$$\begin{aligned} c_b(t) &\approx -\frac{V_{ba}}{2\hbar} \frac{e^{i(\omega_0 - \omega)t/2}}{\omega_0 - \omega} \left[e^{i(\omega_0 - \omega)t/2} - e^{-i(\omega_0 - \omega)t/2} \right] \\ &= -i \frac{V_{ba}}{\hbar} \frac{\sin[(\omega_0 - \omega)t/2]}{\omega_0 - \omega} e^{i(\omega_0 - \omega)t/2}. \end{aligned} \quad (11.34)$$

The **transition probability**—the probability that a particle which started out in the state ψ_a will be found, at time t , in the state ψ_b —is

$$P_{a \rightarrow b}(t) = |c_b(t)|^2 \approx \frac{|V_{ab}|^2}{\hbar^2} \frac{\sin^2[(\omega_0 - \omega)t/2]}{(\omega_0 - \omega)^2}. \quad (11.35)$$

The most remarkable feature of this result is that, as a function of time, the transition probability *oscillates* sinusoidally (Figure 11.4). After rising to a maximum of $|V_{ab}|^2 / \hbar^2 (\omega_0 - \omega)^2$ —necessarily much less than 1, else the assumption that the perturbation is “small” would be invalid—it drops back down to zero! At times $t_n = 2n\pi / |\omega_0 - \omega|$, where $n = 1, 2, 3, \dots$, the particle is *certain* to be back in the lower state. If you want to maximize your chances of provoking a transition, you should *not* keep the perturbation on for a long period; you do better to *turn it off* after a time $\pi / |\omega_0 - \omega|$, and hope to “catch” the system in the upper state. In Problem 11.9 it is shown that this “flopping” is not an artifact of perturbation theory—it occurs also in the exact solution, though the flopping *frequency* is modified somewhat.

As I noted earlier, the probability of a transition is greatest when the driving frequency is close to the “natural” frequency, ω_0 .³ This is illustrated in Figure 11.5, where $P_{a \rightarrow b}$ is plotted as a function of ω . The peak has a height of $(|V_{ab}| t / 2\hbar)^2$ and a width $4\pi/t$; evidently it

³ For very small t , $P_{a \rightarrow b}(t)$ is independent of ω ; it takes a couple of cycles for the system to “realize” that the perturbation is periodic.

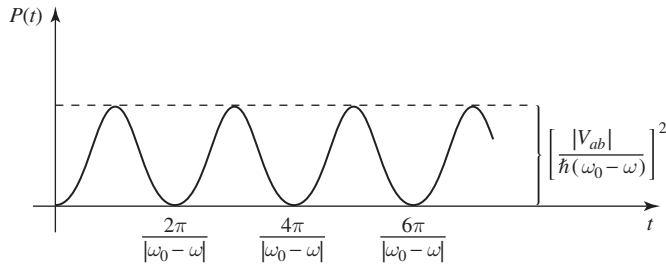


Figure 11.4: Transition probability as a function of time, for a sinusoidal perturbation (Equation 11.35).

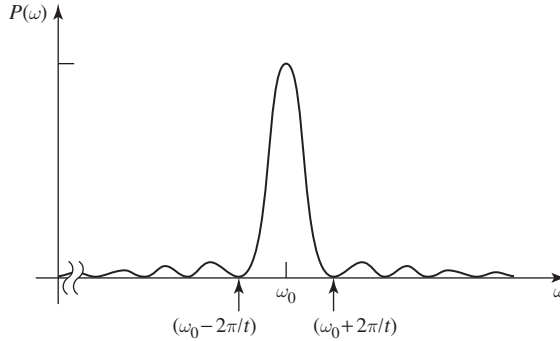


Figure 11.5: Transition probability as a function of driving frequency (Equation 11.35).

gets higher and narrower as time goes on. (Ostensibly, the maximum increases without limit. However, the perturbation assumption breaks down before it gets close to 1, so we can believe the result only for relatively small t . In Problem 11.9 you will see that the *exact* result never exceeds 1.)

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Problem 11.9 The first term in Equation 11.32 comes from the $e^{i\omega t}/2$ part of $\cos(\omega t)$, and the second from $e^{-i\omega t}/2$. Thus dropping the first term is formally equivalent to writing $\hat{H}' = (V/2)e^{-i\omega t}$, which is to say,

$$H'_{ba} = \frac{V_{ba}}{2}e^{-i\omega t}, \quad H'_{ab} = \frac{V_{ab}}{2}e^{i\omega t}. \quad (11.36)$$

(The latter is required to make the Hamiltonian matrix hermitian—or, if you prefer, to pick out the dominant term in the formula analogous to Equation 11.32 for $c_a(t)$.) Rabi noticed that if you make this so-called **rotating wave approximation** at the *beginning* of the calculation, Equation 11.17 can be solved exactly, with no need for perturbation theory, and no assumption about the strength of the field.

- (a) Solve Equation 11.17 in the rotating wave approximation (Equation 11.36), for the usual initial conditions: $c_a(0) = 1$, $c_b(0) = 0$. Express your results ($c_a(t)$ and $c_b(t)$) in terms of the **Rabi flopping frequency**,

$$\omega_r \equiv \frac{1}{2}\sqrt{(\omega - \omega_0)^2 + (|V_{ab}|/\hbar)^2}. \quad (11.37)$$

- (b) Determine the transition probability, $P_{a \rightarrow b}(t)$, and show that it never exceeds 1. Confirm that $|c_a(t)|^2 + |c_b(t)|^2 = 1$.

- (c) Check that $P_{a \rightarrow b}(t)$ reduces to the perturbation theory result (Equation 11.35) when the perturbation is “small,” and state precisely what small *means* in this context, as a constraint on V .
- (d) At what time does the system first return to its initial state?

11.2 EMISSION AND ABSORPTION OF RADIATION

11.2.1 Electromagnetic Waves

An electromagnetic wave (I’ll refer to it as “light”, though it could be infrared, ultraviolet, microwave, x-ray, etc.; these differ only in their frequencies) consists of transverse (and mutually perpendicular) oscillating electric and magnetic fields (Figure 11.6). An atom, in the presence of a passing light wave, responds primarily to the electric component. If the wavelength is long (compared to the size of the atom), we can ignore the *spatial* variation in the field;⁴ the atom, then, is exposed to a sinusoidally oscillating electric field

$$\mathbf{E} = E_0 \cos(\omega t) \hat{k} \quad (11.38)$$

(for the moment I’ll assume the light is monochromatic, and polarized along the z direction). The perturbing Hamiltonian is⁵

$$H' = -q E_0 z \cos(\omega t), \quad (11.39)$$

where q is the charge of the electron.⁶ Evidently⁷

$$H'_{ba} = -\wp E_0 \cos(\omega t), \quad \text{where} \quad \wp \equiv q \langle \psi_b | z | \psi_a \rangle. \quad (11.40)$$

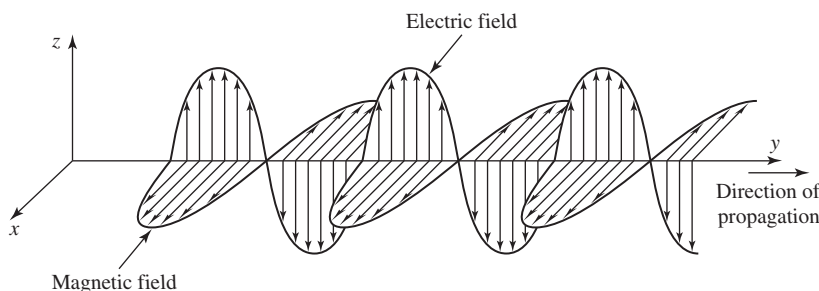


Figure 11.6: An electromagnetic wave.

⁴ For visible light $\lambda \sim 5000 \text{ \AA}$, while the diameter of an atom is around $1 \text{ \AA}$, so this approximation is reasonable; but it would *not* be for x-rays. Problem 11.31 explores the effect of spatial variation of the field.

⁵ The energy of a charge q in a static field $\mathbf{E}$ is $-q \int \mathbf{E} \cdot d\mathbf{r}$. You may well object to the use of an electrostatic formula for a manifestly time-dependent field. I am implicitly assuming that the period of oscillation is long compared to the time it takes the charge to move around (within the atom).

⁶ As usual, we assume the nucleus is heavy and stationary; it is the wave function of the *electron* that concerns us.

⁷ The letter $\wp$ is supposed to remind you of **electric dipole moment** (for which, in electrodynamics, the letter p is customarily used—in this context it is rendered as a squiggly $\wp$ to avoid confusion with momentum). Actually, $\wp$ is the off-diagonal matrix element of the z component of the dipole moment operator, $q\mathbf{r}$. Because of its association with electric dipole moments, radiation governed by Equation 11.40 is called **electric dipole radiation**; it is overwhelmingly the dominant kind, at least in the visible region. See Problem 11.31 for generalizations and terminology.

Typically, ψ is an even or odd function of z ; in either case $z|\psi|^2$ is odd, and integrates to zero (this is Laporte's rule, Section 6.4.3; for some examples see Problem 11.2). This licenses our usual assumption that the diagonal matrix elements of $\hat{H}'$ vanish. Thus the interaction of light with matter is governed by precisely the kind of oscillatory perturbation we studied in Section 11.1.3, with

$$V_{ba} = -\wp E_0. \quad (11.41)$$

11.2.2 Absorption, Stimulated Emission, and Spontaneous Emission

If an atom starts out in the “lower” state ψ_a , and you shine a polarized monochromatic beam of light on it, the probability of a transition to the “upper” state ψ_b is given by Equation 11.35, which (in view of Equation 11.41) takes the form

$$P_{a \rightarrow b}(t) = \left(\frac{|\wp| E_0}{\hbar} \right)^2 \frac{\sin^2[(\omega_0 - \omega)t/2]}{(\omega_0 - \omega)^2}. \quad (11.42)$$

In this process, the atom absorbs energy $E_b - E_a = \hbar\omega_0$ from the electromagnetic field, so it's called **absorption**. (Informally, we say that the atom has “absorbed a photon” (Figure 11.7(a).) Technically, the word “photon” belongs to **quantum electrodynamics**—the quantum theory of the electromagnetic field—whereas we are treating the field itself *classically*. But this language is convenient, as long as you don't read too much into it.)

I could, of course, go back and run the whole derivation for a system that starts off in the *upper* state ($c_a(0) = 0$, $c_b(0) = 1$). Do it for yourself, if you like; it comes out *exactly the same*—except that this time we're calculating $P_{b \rightarrow a} = |c_a(t)|^2$, the probability of a transition *down* to the *lower* level:

$$P_{b \rightarrow a}(t) = \left(\frac{|\wp| E_0}{\hbar} \right)^2 \frac{\sin^2[(\omega_0 - \omega)t/2]}{(\omega_0 - \omega)^2}. \quad (11.43)$$

(It *has* to come out this way—all we're doing is switching $a \leftrightarrow b$, which substitutes $-\omega_0$ for ω_0 . When we get to Equation 11.32 we now keep the *first* term, with $-\omega_0 + \omega$ in the denominator, and the rest is the same as before.) But when you stop to think of it, this is an absolutely *astounding* result: If the particle is in the *upper* state, and you shine light on it, it can make a transition to the *lower* state, and in fact the probability of such a transition is exactly the same as for a transition *upward* from the *lower* state. This process, which was first predicted by Einstein, is called **stimulated emission**.

In the case of stimulated emission the electromagnetic field *gains* energy $\hbar\omega_0$ from the atom; we say that one photon went in and *two* photons came out—the original one that caused the transition plus another one from the transition itself (Figure 11.7(b)). This raises the possibility of *amplification*, for if I had a bottle of atoms, all in the upper state, and triggered it with a single incident photon, a chain reaction would occur, with the first photon producing two, these two producing four, and so on. We'd have an enormous number of photons coming out, all with the same frequency and at virtually the same instant. This is the principle behind the **laser** (light

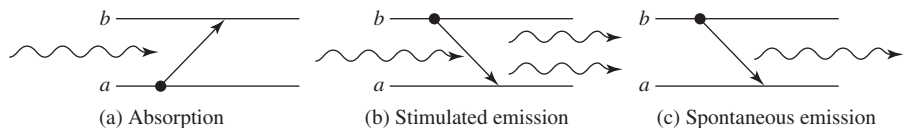


Figure 11.7: Three ways in which light interacts with atoms: (a) absorption, (b) stimulated emission, (c) spontaneous emission.

amplification by stimulated emission of radiation). Note that it is essential (for laser action) to get a majority of the atoms into the upper state (a so-called **population inversion**), because *absorption* (which *costs* one photon) competes with stimulated emission (which *creates* one); if you started with an even mixture of the two states, you'd get no amplification at all.

There is a *third* mechanism (in addition to absorption and stimulated emission) by which radiation interacts with matter; it is called **spontaneous emission**. Here an atom in the excited state makes a transition downward, with the release of a photon, but *without* any applied electromagnetic field to initiate the process (Figure 11.7(c)). This is the mechanism that accounts for the normal decay of an atomic excited state. At first sight it is far from clear why spontaneous emission should occur at *all*. If the atom is in a stationary state (albeit an excited one), and there is no external perturbation, it should just sit there forever. And so it *would*, if it were *really* free of all external perturbations. However, in quantum electrodynamics the fields are nonzero *even in the ground state*—just as the harmonic oscillator (for example) has nonzero energy (to wit: $\hbar\omega/2$) in its ground state. You can turn out all the lights, and cool the room down to absolute zero, but there is still some electromagnetic radiation present, and it is this “zero point” radiation that serves to catalyze spontaneous emission. When you come right down to it, there is really no such thing as *truly* spontaneous emission; it's *all* stimulated emission. The only distinction to be made is whether the field that does the stimulating is one that *you* put there, or one that *God* put there. In this sense it is exactly the reverse of the classical radiative process, in which it's *all* spontaneous, and there is no such thing as *stimulated* emission.

Quantum electrodynamics is beyond the scope of this book,⁸ but there is a lovely argument, due to Einstein,⁹ which interrelates the three processes (absorption, stimulated emission, and spontaneous emission). Einstein did not identify the *mechanism* responsible for spontaneous emission (perturbation by the ground-state electromagnetic field), but his results nevertheless enable us to calculate the spontaneous emission rate, and from that the natural lifetime of an excited atomic state.¹⁰ Before we turn to that, however, we need to consider the response of an atom to non-monochromatic, unpolarized, incoherent electromagnetic waves coming in from all directions—such as it would encounter, for instance, if it were immersed in thermal radiation.

11.2.3 Incoherent Perturbations

The energy density in an electromagnetic wave is¹¹

$$u = \frac{\epsilon_0}{2} E_0^2, \quad (11.44)$$

⁸ For an accessible treatment see Rodney Loudon, *The Quantum Theory of Light*, 2nd edn (Clarendon Press, Oxford, 1983).

⁹ Einstein's paper was published in 1917, well before the Schrödinger equation. Quantum electrodynamics comes into the argument via the Planck blackbody formula, which dates from 1900.

¹⁰ For an alternative derivation using “seat-of-the-pants” quantum electrodynamics, see Problem 11.11.

¹¹ David J. Griffiths, *Introduction to Electrodynamics*, 4th edn, (Cambridge University Press, Cambridge, UK, 2017), Section 9.2.3. In general, the energy per unit volume in electromagnetic fields is

$$u = (\epsilon_0/2) E^2 + (1/2\mu_0) B^2.$$

For electromagnetic waves, the electric and magnetic contributions are equal, so

$$u = \epsilon_0 E^2 = \epsilon_0 E_0^2 \cos^2(\omega t),$$

and the average over a full cycle is $(\epsilon_0/2) E_0^2$, since the average of $\cos^2$ (or $\sin^2$) is 1/2.

where E_0 is (as before) the amplitude of the electric field. So the transition probability (Equation 11.43) is (not surprisingly) proportional to the energy density of the fields:

$$P_{b \rightarrow a}(t) = \frac{2u}{\epsilon_0 \hbar^2} |\wp|^2 \frac{\sin^2[(\omega_0 - \omega)t/2]}{(\omega_0 - \omega)^2}. \quad (11.45)$$

But this is for a **monochromatic** wave, at a single frequency ω . In many applications the system is exposed to electromagnetic waves at a whole *range* of frequencies; in that case $u \rightarrow \rho(\omega)d\omega$, where $\rho(\omega)d\omega$ is the energy density in the frequency range $d\omega$, and the net transition probability takes the form of an integral:¹²

$$P_{b \rightarrow a}(t) = \frac{2}{\epsilon_0 \hbar^2} |\wp|^2 \int_0^\infty \rho(\omega) \left\{ \frac{\sin^2[(\omega_0 - \omega)t/2]}{(\omega_0 - \omega)^2} \right\} d\omega. \quad (11.46)$$

The term in curly brackets is sharply peaked about ω_0 (Figure 11.5), whereas $\rho(\omega)$ is ordinarily quite broad, so we may as well replace $\rho(\omega)$ by $\rho(\omega_0)$, and take it outside the integral:

$$P_{b \rightarrow a}(t) \approx \frac{2|\wp|^2}{\epsilon_0 \hbar^2} \rho(\omega_0) \int_0^\infty \frac{\sin^2[(\omega_0 - \omega)t/2]}{(\omega_0 - \omega)^2} d\omega. \quad (11.47)$$

Changing variables to $x \equiv (\omega_0 - \omega)t/2$, extending the limits of integration to $x = \pm\infty$ (since the integrand is essentially zero out there anyway), and looking up the definite integral

$$\int_{-\infty}^\infty \frac{\sin^2 x}{x^2} dx = \pi, \quad (11.48)$$

we find

$$P_{b \rightarrow a}(t) \approx \frac{\pi |\wp|^2}{\epsilon_0 \hbar^2} \rho(\omega_0) t. \quad (11.49)$$

This time the transition probability is proportional to t . The bizarre “flopping” phenomenon characteristic of a monochromatic perturbation gets “washed out” when we hit the system with an incoherent spread of frequencies. In particular, the **transition rate** ($R \equiv dP/dt$) is now a *constant*:

$$R_{b \rightarrow a} = \frac{\pi}{\epsilon_0 \hbar^2} |\wp|^2 \rho(\omega_0). \quad (11.50)$$

Up to now, we have assumed that the perturbing wave is coming in along the y direction (Figure 11.6), and polarized in the z direction. But we are interested in the case of an atom bathed in radiation coming from *all* directions, and with all possible polarizations; the energy in the fields ($\rho(\omega)$) is shared equally among these different modes. What we need, in place of $|\wp|^2$, is the *average* of $|\wp \cdot \hat{n}|^2$, where

$$\wp \equiv q \langle \psi_b | \mathbf{r} | \psi_a \rangle \quad (11.51)$$

(generalizing Equation 11.40), and the average is over all polarizations and all incident directions.

¹² Equation 11.46 assumes that the perturbations at different frequencies are *independent*, so that the total transition probability is a sum of the individual probabilities. If the different components are **coherent** (phase-correlated), then we should add *amplitudes* ($c_b(t)$), not *probabilities* ($|c_b(t)|^2$), and there will be cross-terms. For the applications we will consider the perturbations are always incoherent.

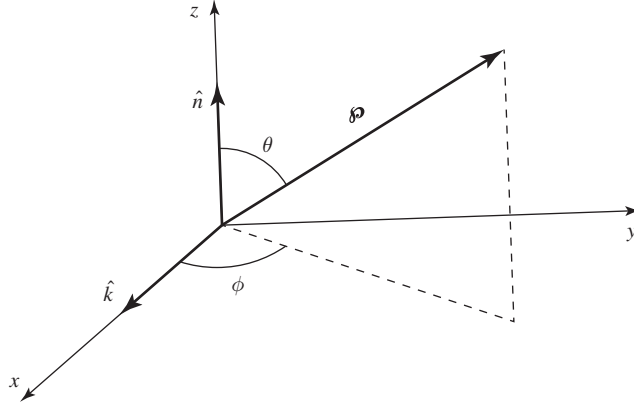


Figure 11.8: Axes for the averaging of $|\boldsymbol{\rho} \cdot \hat{n}|^2$.

The averaging can be carried out as follows: Choose spherical coordinates such that the direction of propagation ($\hat{k}$) is along x , the polarization ($\hat{n}$) is along z , and the vector $\boldsymbol{\rho}$ defines the spherical angles θ and ϕ (Figure 11.8).¹³ (Actually, $\boldsymbol{\rho}$ is *fixed*, here, and we're averaging over all $\hat{k}$ and $\hat{n}$ consistent with $\hat{k} \perp \hat{n}$ —which is to say, over all θ and ϕ . But we might as well integrate over all directions of $\boldsymbol{\rho}$, keeping $\hat{k}$ and $\hat{n}$ fixed—it amounts to the same thing.) Then

$$\boldsymbol{\rho} \cdot \hat{n} = \rho \cos \theta, \quad (11.52)$$

and

$$\begin{aligned} |\boldsymbol{\rho} \cdot \hat{n}|_{\text{ave}}^2 &= \frac{1}{4\pi} \int |\boldsymbol{\rho}|^2 \cos^2 \theta \sin \theta \, d\theta \, d\phi \\ &= \frac{|\boldsymbol{\rho}|^2}{4\pi} \left(-\frac{\cos^3 \theta}{3} \right) \Big|_0^\pi (2\pi) = \frac{1}{3} |\boldsymbol{\rho}|^2. \end{aligned} \quad (11.53)$$

Conclusion: The transition rate for stimulated emission from state b to state a , under the influence of incoherent, unpolarized light incident from all directions, is

$$R_{b \rightarrow a} = \frac{\pi}{3\epsilon_0 \hbar^2} |\boldsymbol{\rho}|^2 \rho(\omega_0), \quad (11.54)$$

where $\boldsymbol{\rho}$ is the matrix element of the electric dipole moment between the two states (Equation 11.51), and $\rho(\omega_0)$ is the energy density in the fields, per unit frequency, evaluated at $\omega_0 = (E_b - E_a)/\hbar$.

¹³ I'll treat $\boldsymbol{\rho}$ as though it were *real*, even though in general it will be complex. Since

$$|\boldsymbol{\rho} \cdot \hat{n}|^2 = |\text{Re}(\boldsymbol{\rho}) \cdot \hat{n} + i \text{Im}(\boldsymbol{\rho}) \cdot \hat{n}|^2 = |\text{Re}(\boldsymbol{\rho}) \cdot \hat{n}|^2 + |\text{Im}(\boldsymbol{\rho}) \cdot \hat{n}|^2$$

we can do the whole calculation for the real and imaginary parts separately, and simply add the results. In Equation 11.54 the absolute value signs denote *both* the vector magnitude *and* the complex amplitude:

$$|\boldsymbol{\rho}|^2 = |\rho_x|^2 + |\rho_y|^2 + |\rho_z|^2.$$

11.3 SPONTANEOUS EMISSION

11.3.1 Einstein's A and B Coefficients

Picture a container of atoms, N_a of them in the lower state (ψ_a), and N_b of them in the upper state (ψ_b). Let A be the spontaneous emission rate,¹⁴ so that the number of particles leaving the upper state by this process, per unit time, is $N_b A$.¹⁵ The transition rate for stimulated emission, as we have seen (Equation 11.54), is proportional to the energy density of the electromagnetic field: $B_{ba}\rho(\omega_0)$, where $B_{ba} = \pi |\mathbf{p}|^2 / 3\epsilon_0 \hbar^2$; the number of particles leaving the upper state by this mechanism, per unit time, is $N_b B_{ba}\rho(\omega_0)$. The absorption rate is likewise proportional to $\rho(\omega_0)$ —call it $B_{ab}\rho(\omega_0)$; the number of particles per unit time *joining* the upper level is therefore $N_a B_{ab}\rho(\omega_0)$. All told, then,

$$\frac{dN_b}{dt} = -N_b A - N_b B_{ba}\rho(\omega_0) + N_a B_{ab}\rho(\omega_0). \quad (11.55)$$

Suppose these atoms are in thermal equilibrium with the ambient field, so that the number of particles in each level is *constant*. In that case $dN_b/dt = 0$, and it follows that

$$\rho(\omega_0) = \frac{A}{(N_a/N_b) B_{ab} - B_{ba}}. \quad (11.56)$$

On the other hand, we know from statistical mechanics¹⁶ that the number of particles with energy E , in thermal equilibrium at temperature T , is proportional to the **Boltzmann factor**, $\exp(-E/k_B T)$, so

$$\frac{N_a}{N_b} = \frac{e^{-E_a/k_B T}}{e^{-E_b/k_B T}} = e^{\hbar\omega_0/k_B T}, \quad (11.57)$$

and hence

$$\rho(\omega_0) = \frac{A}{e^{\hbar\omega_0/k_B T} B_{ab} - B_{ba}}. \quad (11.58)$$

But Planck's blackbody formula¹⁷ tells us the energy density of thermal radiation:

$$\rho(\omega) = \frac{\hbar}{\pi^2 c^3} \frac{\omega^3}{e^{\hbar\omega/k_B T} - 1}; \quad (11.59)$$

comparing the two expressions, we conclude that

$$B_{ab} = B_{ba} \quad (11.60)$$

and

$$A = \frac{\omega_0^3 \hbar}{\pi^2 c^3} B_{ba}. \quad (11.61)$$

¹⁴ Normally I'd use R for a transition rate, but out of deference to *der Alte* everyone follows Einstein's notation in this context.

¹⁵ Assume that N_a and N_b are very large, so we can treat them as continuous functions of time and ignore statistical fluctuations.

¹⁶ See, for example, Daniel Schroeder, *An Introduction to Thermal Physics* (Pearson, Upper Saddle River, NJ, 2000), Section 6.1.

¹⁷ Schroeder, footnote 16, Section 7.4.

Equation 11.60 confirms what we already knew: the transition rate for stimulated emission is the same as for absorption. But it was an astonishing result in 1917—indeed, Einstein was forced to “invent” stimulated emission in order to reproduce Planck’s formula. Our present attention, however, focuses on Equation 11.61, for this tells us the spontaneous emission rate (A)—which is what we are looking for—in terms of the stimulated emission rate ($B_{ba}\rho(\omega_0)$)—which we already know. From Equation 11.54 we read off

$$B_{ba} = \frac{\pi}{3\epsilon_0\hbar^2} |\mathbf{p}|^2, \quad (11.62)$$

and it follows that the spontaneous emission rate is

$$A = \frac{\omega_0^3 |\mathbf{p}|^2}{3\pi\epsilon_0\hbar c^3}. \quad (11.63)$$

Problem 11.10 As a mechanism for downward transitions, spontaneous emission competes with thermally stimulated emission (stimulated emission for which blackbody radiation is the source). Show that at room temperature ($T = 300$ K) thermal stimulation dominates for frequencies well below 5×10^{12} Hz, whereas spontaneous emission dominates for frequencies well above 5×10^{12} Hz. Which mechanism dominates for visible light?

Problem 11.11 You could derive the spontaneous emission rate (Equation 11.63) without the detour through Einstein’s A and B coefficients if you knew the ground state energy density of the electromagnetic field, $\rho_0(\omega)$, for then it would simply be a case of stimulated emission (Equation 11.54). To do this honestly would require quantum electrodynamics, but if you are prepared to believe that the ground state consists of *one photon in each classical mode*, then the derivation is fairly simple:

- (a) To obtain the classical modes, consider an empty cubical box, of side l , with one corner at the origin. Electromagnetic fields (in vacuum) satisfy the classical wave equation¹⁸

$$\left(\frac{1}{c^2} \frac{\partial^2}{\partial t^2} - \nabla^2 \right) f(x, y, z, t) = 0,$$

where f stands for any component of $\mathbf{E}$ or of $\mathbf{B}$. Show that separation of variables, and the imposition of the boundary condition $f = 0$ on all six surfaces yields the standing wave patterns

$$f_{n_x, n_y, n_z} = A \cos(\omega t) \sin\left(\frac{n_x \pi}{l} x\right) \sin\left(\frac{n_y \pi}{l} y\right) \sin\left(\frac{n_z \pi}{l} z\right),$$

with

$$\omega = \frac{\pi c}{l} \sqrt{n_x^2 + n_y^2 + n_z^2}.$$

There are two modes for each triplet of positive integers ($n_x, n_y, n_z = 1, 2, 3, \dots$), corresponding to the two polarization states.

¹⁸ Griffiths, footnote 11, Section 9.2.1.

- (b) The energy of a photon is $E = h\nu = \hbar\omega$ (Equation 4.92), so the energy in the mode (n_x, n_y, n_z) is

$$E_{n_x, n_y, n_z} = 2 \frac{\pi \hbar c}{l} \sqrt{n_x^2 + n_y^2 + n_z^2}.$$

What, then, is the *total* energy per unit volume in the frequency range $d\omega$, if each mode gets one photon? Express your answer in the form

$$\frac{1}{l^3} dE = \rho_0(\omega) d\omega$$

and read off $\rho_0(\omega)$. *Hint:* refer to Figure 5.3.

- (c) Use your result, together with Equation 11.54, to obtain the spontaneous emission rate. Compare Equation 11.63.

11.3.2 The Lifetime of an Excited State

Equation 11.63 is our fundamental result; it gives the transition rate for spontaneous emission. Suppose, now, that you have somehow pumped a large number of atoms into the excited state. As a result of spontaneous emission, this number will decrease as time goes on; specifically, in a time interval dt you will lose a fraction $A dt$ of them:

$$dN_b = -AN_b dt, \quad (11.64)$$

(assuming there is no mechanism to replenish the supply).¹⁹ Solving for $N_b(t)$, we find:

$$N_b(t) = N_b(0)e^{-At}; \quad (11.65)$$

evidently the number remaining in the excited state decreases exponentially, with a time constant

$$\tau = \frac{1}{A}. \quad (11.66)$$

We call this the **lifetime** of the state—technically, it is the time it takes for $N_b(t)$ to reach $1/e \approx 0.368$ of its initial value.

I have assumed all along that there are only *two* states for the system, but this was just for notational simplicity—the spontaneous emission formula (Equation 11.63) gives the transition rate for $\psi_b \rightarrow \psi_a$ regardless of what other states may be accessible (see Problem 11.24). Typically, an excited atom has many different **decay modes** (that is: ψ_b can decay to a large number of different lower-energy states, $\psi_{a_1}, \psi_{a_2}, \psi_{a_3}, \dots$). In that case the transition rates *add*, and the net lifetime is

$$\tau = \frac{1}{A_1 + A_2 + A_3 + \dots}. \quad (11.67)$$

¹⁹ This situation is not to be confused with the case of thermal equilibrium, which we considered in the previous section. We assume here that the atoms have been lifted *out* of equilibrium, and are in the process of cascading back down to their equilibrium levels.

Example 11.1

Suppose a charge q is attached to a spring and constrained to oscillate along the x axis. Say it starts out in the state $|n\rangle$ (Equation 2.68), and decays by spontaneous emission to state $|n'\rangle$. From Equation 11.51 we have

$$\mathbf{p} = q \langle n | x | n' \rangle \hat{i}.$$

You calculated the matrix elements of x back in Problem 3.39:

$$\langle n | x | n' \rangle = \sqrt{\frac{\hbar}{2m\omega}} \left(\sqrt{n'} \delta_{n,n'-1} + \sqrt{n} \delta_{n',n-1} \right),$$

where ω is the natural frequency of the oscillator (I no longer need this letter for the frequency of the stimulating radiation). But we're talking about *emission*, so n' must be *lower* than n ; for our purposes, then,

$$\mathbf{p} = q \sqrt{\frac{n\hbar}{2m\omega}} \delta_{n',n-1} \hat{i}. \quad (11.68)$$

Evidently transitions occur only to states one step lower on the “ladder”, and the frequency of the photon emitted is

$$\omega_0 = \frac{E_n - E_{n'}}{\hbar} = \frac{(n + 1/2)\hbar\omega - (n' + 1/2)\hbar\omega}{\hbar} = (n - n')\omega = \omega. \quad (11.69)$$

Not surprisingly, the system radiates at the classical oscillator frequency. The transition rate (Equation 11.63) is

$$A = \frac{nq^2\omega^2}{6\pi\epsilon_0 mc^3}, \quad (11.70)$$

and the lifetime of the n th stationary state is

$$\tau_n = \frac{6\pi\epsilon_0 mc^3}{nq^2\omega^2}. \quad (11.71)$$

Meanwhile, each radiated photon carries an energy $\hbar\omega$, so the *power* radiated is $A\hbar\omega$:

$$P = \frac{q^2\omega^2}{6\pi\epsilon_0 mc^3} (n\hbar\omega),$$

or, since the energy of an oscillator in the n th state is $E = (n + 1/2)\hbar\omega$,

$$P = \frac{q^2\omega^2}{6\pi\epsilon_0 mc^3} \left(E - \frac{1}{2}\hbar\omega \right). \quad (11.72)$$

This is the average power radiated by a quantum oscillator with (initial) energy E .

For comparison, let's determine the average power radiated by a *classical* oscillator with the same energy. According to classical electrodynamics, the power radiated by an accelerating charge q is given by the **Larmor formula**:²⁰

$$P = \frac{q^2 a^2}{6\pi\epsilon_0 c^3}. \quad (11.73)$$

²⁰ See, for example, Griffiths, footnote 11, Section 11.2.1.

For a harmonic oscillator with amplitude x_0 , $x(t) = x_0 \cos(\omega t)$, and the acceleration is $a = -x_0 \omega^2 \cos(\omega t)$. Averaging over a full cycle, then,

$$P = \frac{q^2 x_0^2 \omega^4}{12\pi \epsilon_0 c^3}.$$

But the *energy* of the oscillator is $E = (1/2) m \omega^2 x_0^2$, so $x_0^2 = 2E/m\omega^2$, and hence

$$P = \frac{q^2 \omega^2}{6\pi \epsilon_0 m c^3} E. \quad (11.74)$$

This is the average power radiated by a *classical* oscillator with energy E . In the classical limit ($\hbar \rightarrow 0$) the classical and quantum formulas agree;²¹ however, the quantum formula (Equation 11.72) protects the ground state: If $E = (1/2) \hbar \omega$ the oscillator does not radiate.

Problem 11.12 The **half-life** ($t_{1/2}$) of an excited state is the time it would take for half the atoms in a large sample to make a transition. Find the relation between $t_{1/2}$ and τ (the “lifetime” of the state).

* **Problem 11.13** Calculate the lifetime (in seconds) for each of the four $n = 2$ states of hydrogen. *Hint:* You’ll need to evaluate matrix elements of the form $\langle \psi_{100} | x | \psi_{200} \rangle$, $\langle \psi_{100} | y | \psi_{211} \rangle$, and so on. Remember that $x = r \sin \theta \cos \phi$, $y = r \sin \theta \sin \phi$, and $z = r \cos \theta$. Most of these integrals are zero, so inspect them closely before you start calculating. *Answer:* 1.60×10^{-9} seconds for all except ψ_{200} , which is infinite.

11.3.3 Selection Rules

The calculation of spontaneous emission rates has been reduced to a matter of evaluating matrix elements of the form

$$\langle \psi_b | \mathbf{r} | \psi_a \rangle.$$

As you will have discovered if you worked Problem 11.13, (if you *didn’t*, go back right now and *do* so!) these quantities are very often *zero*, and it would be helpful to know in advance when this is going to happen, so we don’t waste a lot of time evaluating unnecessary integrals. Suppose we are interested in systems like hydrogen, for which the Hamiltonian is spherically symmetrical. In that case we can specify the states with the usual quantum numbers n , ℓ , and m , and the matrix elements are

$$\langle n' \ell' m' | \mathbf{r} | n \ell m \rangle.$$

Now, $\mathbf{r}$ is a vector operator, and we can invoke the results of Chapter 6 to obtain the **selection rules**²²

$$\Delta \ell \equiv \ell' - \ell = \pm 1, \quad \Delta m \equiv m' - m = 0 \text{ or } \pm 1. \quad (11.75)$$

²¹ This is an example of Bohr’s **Correspondence Principle**. In fact, if we express P in terms of the energy *above the ground state*, the two formulas are identical.

²² See Equation 6.62 (Equation 6.26 eliminates $\Delta \ell = 0$), or derive them from scratch using Problems 11.14 and 11.15.

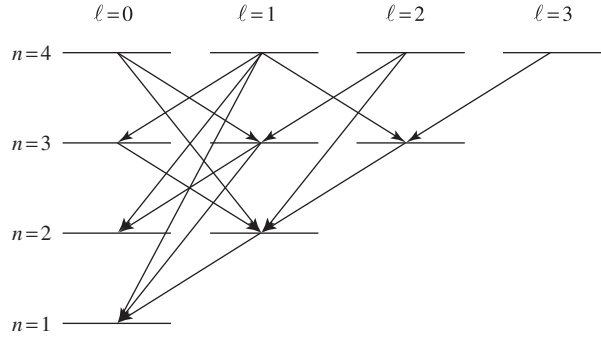


Figure 11.9: Allowed decays for the first four Bohr levels in hydrogen.

These conditions follow from symmetry alone. If they are *not* met, then the matrix element is zero, and the transition is said to be **forbidden**. Moreover, it follows from Equations 6.56–6.58 that

$$\begin{cases} \text{if } m' = m, & \text{then } \langle n' \ell' m' | x | n \ell m \rangle = \langle n' \ell' m' | y | n \ell m \rangle = 0, \\ \text{if } m' = m \pm 1, & \text{then } \langle n' \ell' m' | x | n \ell m \rangle = \pm i \langle n' \ell' m' | y | n \ell m \rangle \\ & \text{and } \langle n' \ell' m' | z | n \ell m \rangle = 0. \end{cases} \quad (11.76)$$

So it is never necessary to compute the matrix elements of both x and y ; you can always get one from the other.

Evidently not all transitions to lower-energy states can proceed by electric dipole radiation; most are forbidden by the selection rules. The scheme of allowed transitions for the first four Bohr levels in hydrogen is shown in Figure 11.9. Notice that the $2S$ state (ψ_{200}) is “stuck”: it cannot decay, because there is no lower-energy state with $\ell = 1$. It is called a **metastable** state, and its lifetime is indeed much longer than that of, for example, the $2P$ states (ψ_{211} , ψ_{210} , and ψ_{21-1}). Metastable states do eventually decay, by collisions, or by “forbidden” transitions (Problem 11.31), or by multiphoton emission.

Problem 11.14 From the commutators of L_z with x , y , and z (Equation 4.122):

$$[L_z, x] = i\hbar y, \quad [L_z, y] = -i\hbar x, \quad [L_z, z] = 0, \quad (11.77)$$

obtain the selection rule for Δm and Equation 11.76. *Hint:* Sandwich each commutator between $\langle n' \ell' m' |$ and $| n \ell m \rangle$.

**

Problem 11.15 Obtain the selection rule for $\Delta \ell$ as follows:

(a) Derive the commutation relation

$$[L^2, [L^2, \mathbf{r}]] = 2\hbar^2 (\mathbf{r}L^2 + L^2\mathbf{r}). \quad (11.78)$$

Hint: First show that

$$[L^2, z] = 2i\hbar (xL_y - yL_x - i\hbar z).$$

Use this, and (in the final step) the fact that $\mathbf{r} \cdot \mathbf{L} = \mathbf{r} \cdot (\mathbf{r} \times \mathbf{p}) = 0$, to demonstrate that

$$[L^2, [L^2, z]] = 2\hbar^2 (zL^2 + L^2z).$$

The generalization from z to $\mathbf{r}$ is trivial.

(b) Sandwich this commutator between $\langle n' \ell' m' |$ and $|n \ell m\rangle$, and work out the implications.

**

Problem 11.16 An electron in the $n = 3, \ell = 0, m = 0$ state of hydrogen decays by a sequence of (electric dipole) transitions to the ground state.

(a) What decay routes are open to it? Specify them in the following way:

$$|300\rangle \rightarrow |n \ell m\rangle \rightarrow |n' \ell' m'\rangle \rightarrow \cdots \rightarrow |100\rangle.$$

(b) If you had a bottle full of atoms in this state, what fraction of them would decay via each route?

(c) What is the lifetime of this state? *Hint:* Once it's made the first transition, it's no longer in the state $|300\rangle$, so only the first step in each sequence is relevant in computing the lifetime.

11.4 FERMİ'S GOLDEN RULE

In the previous sections we considered transitions between two *discrete* energy states, such as two bound states of an atom. We saw that such a transition was most likely when the final energy satisfied the resonance condition: $E_f = E_i + \hbar \omega$, where ω is the frequency associated with the perturbation. I now want to look at the case where E_f falls in a *continuum* of states (Figure 11.10). To stick close to the example of Section 11.2, if the radiation is energetic enough it can ionize the atom—the **photoelectric effect**—exciting the electron from a bound state into the continuum of scattering states.

We can't talk about a transition to a *precise* state in that continuum (any more than we can talk about someone being *precisely* 16 years old), but we can compute the probability that the system makes a transition to a state with an energy in some finite range ΔE about E_f . That is given by the integral of Equation 11.35 over all the final states:

$$P = \int_{E_f - \Delta E/2}^{E_f + \Delta E/2} \frac{|V_{in}|^2}{\hbar^2} \left\{ \frac{\sin^2[(\omega_0 - \omega) t/2]}{(\omega_0 - \omega)^2} \right\} \rho(E_n) dE_n, \quad (11.79)$$

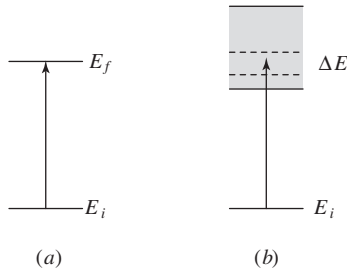


Figure 11.10: A transition (a) between two discrete states and (b) between a discrete state and a continuum of states.

where $\omega_0 = (E_n - E_i) / \hbar$. The quantity $\rho(E) dE$ is the number of states with energy between E and $E + dE$; $\rho(E)$ is called the **density of states**, and I'll show you how it's calculated in Example 11.2.

At short times, Equation 11.79 leads to a transition probability proportional to t^2 , just as for a transition between discrete states. On the other hand, at long times the quantity in curly brackets in Equation 11.79 is sharply peaked: as a function of E_n its maximum occurs at $E_f = E_i + \hbar \omega$ and the central peak has a width of $4\pi \hbar/t$. For sufficiently large t , we can therefore approximate Equation 11.79 as²³

$$P = \frac{|V_{if}|^2}{\hbar^2} \rho(E_f) \int_{-\infty}^{\infty} \frac{\sin^2[(\omega_0 - \omega)t/2]}{(\omega_0 - \omega)^2} dE_n.$$

The remaining integral was already evaluated in Section 11.2.3:

$$P = \frac{2\pi}{\hbar} \left| \frac{V_{if}}{2} \right|^2 \rho(E_f) t. \quad (11.80)$$

The oscillatory behavior of P has again been “washed out,” giving a constant transition rate:²⁴

$$R = \frac{2\pi}{\hbar} \left| \frac{V_{if}}{2} \right|^2 \rho(E_f). \quad (11.81)$$

Equation 11.81 is known as **Fermi's Golden Rule**.²⁵ Apart from the factor of $2\pi/\hbar$, it says that the transition rate is the square of the matrix element (this encapsulates all the relevant information about the *dynamics* of the process) times the density of states (how many final states are accessible, given the energy supplied by the perturbation—the more roads are open, the faster the traffic will flow). It makes sense.

²³ This is the same set of approximations we made in Equations 11.46–11.48.

²⁴ In deriving Equation 11.35, our perturbation was

$$\hat{H}' = V \cos(\omega t) \rightarrow \frac{V}{2} e^{-i\omega t}$$

since we dropped the other (off-resonance) exponential. That is the source of the two inside the absolute value in Equation 11.81. Fermi's Golden rule can also be applied to a constant perturbation, $\hat{H}' = \hat{V}$, if we set $\omega = 0$ drop the 2:

$$R = \frac{2\pi}{\hbar} |V_{if}|^2 \rho(E_f).$$

²⁵ It is actually due to Dirac, but Fermi is the one who gave it the memorable name. See T. Visser, *Am. J. Phys.* **77**, 487 (2009) for the history. Fermi's Golden Rule doesn't just apply to transitions to a continuum of states. For instance, Equation 11.54 can be considered an example. In that case, we integrated over a continuous range of *perturbation frequencies*—not a continuum of final states—but the end result is the same.

Example 11.2

Use Fermi's Golden Rule to obtain the differential scattering cross-section for a particle of mass m and incident wave vector $\mathbf{k}'$ scattering from a potential $V(\mathbf{r})$ (Figure 11.11).

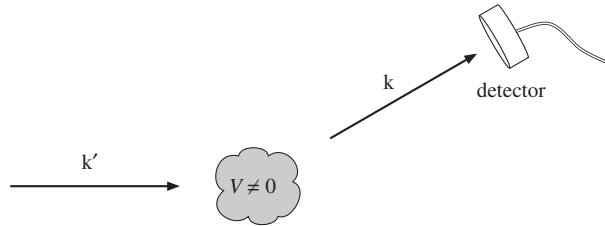


Figure 11.11: A particle with incident wave vector $\mathbf{k}'$ is scattered into a state with wave vector $\mathbf{k}$.

Solution:

We take our initial and final states to be plane waves:

$$\psi_i = \frac{1}{\sqrt{l^3}} e^{i\mathbf{k}' \cdot \mathbf{r}} \text{ and } \psi_f = \frac{1}{\sqrt{l^3}} e^{i\mathbf{k} \cdot \mathbf{r}}. \quad (11.82)$$

Here I've used a technique called **box normalization**; I place the whole setup inside a box of length l on a side. This makes the free-particle states normalizable and countable. Formally, we want the limit $l \rightarrow \infty$; in practice l will drop out of our final expression. Using periodic boundary conditions,²⁶ the allowed values of $\mathbf{k}$ are

$$\mathbf{k} = \frac{2\pi}{l} \left(n_x \hat{i} + n_y \hat{j} + n_z \hat{k} \right) \quad (11.83)$$

for integers n_x, n_y , and n_z . Our perturbation is the scattering potential, $\hat{H}' = V(\mathbf{r})$, and the relevant matrix element is

$$V_{fi} = \int \psi_f^*(\mathbf{r}) V(\mathbf{r}) \psi_i(\mathbf{r}) d^3\mathbf{r} = \frac{1}{l^3} \int e^{i(\mathbf{k}' - \mathbf{k}) \cdot \mathbf{r}} V(\mathbf{r}) d^3\mathbf{r}. \quad (11.84)$$

We need to determine the density of states. In a scattering experiment we measure the number of particles scattered into a solid angle $d\Omega$. We want to count the number of states with energies between E and $E + dE$, with wave vectors $\mathbf{k}$ lying inside $d\Omega$. In k space these states occupy a section of a spherical shell of radius k and thickness dk that subtends a solid angle $d\Omega$; it has a volume

$$k^2 dk d\Omega$$

and contains a number of states²⁷

$$\rho(E) dE = \frac{k^2 dk d\Omega}{(2\pi/l)^3} = \left(\frac{l}{2\pi} \right)^3 k^2 \frac{dk}{dE} dE d\Omega.$$

²⁶ Periodic boundary conditions are discussed in Problem 5.39. In the present context we use periodic boundary conditions—as opposed to impenetrable walls—because they admit traveling-wave solutions.

²⁷ Each state in k -space “occupies” a volume of $(2\pi/l)^3$, as shown in Problem 5.39.

Since $E = \hbar^2 k^2 / 2m$ this gives

$$\rho(E) = \left(\frac{l}{2\pi} \right)^3 \frac{\sqrt{2m^3 E}}{\hbar^3} d\Omega. \quad (11.85)$$

From Fermi's Golden Rule, the rate at which particles are scattered into the solid angle $d\Omega$ is²⁸

$$R_{i \rightarrow d\Omega} = \frac{2\pi}{\hbar} \frac{1}{l^6} \left| \int e^{i(\mathbf{k}' - \mathbf{k}) \cdot \mathbf{r}} V(\mathbf{r}) d^3\mathbf{r} \right|^2 \left(\frac{l}{2\pi} \right)^3 \frac{\sqrt{2m^3 E_f}}{\hbar^3} d\Omega.$$

This is closely related to the differential scattering cross section:

$$\frac{d\sigma}{d\Omega} = \frac{R_{i \rightarrow d\Omega}}{J_i d\Omega} \quad (11.86)$$

where J_i is the flux (or probability current) of incident particles. For an incident wave of the form $\psi_i = A e^{i\mathbf{k}' \cdot \mathbf{r}'}$, the probability current is (Equation 4.220).

$$J_i = |A|^2 v = \frac{1}{l^3} \frac{\hbar k'}{m} \quad (11.87)$$

and

$$\frac{d\sigma}{d\Omega} = \left| -\frac{m}{2\pi\hbar^2} \int e^{i(\mathbf{k} - \mathbf{k}') \cdot \mathbf{r}} V(\mathbf{r}) d^3\mathbf{r} \right|^2. \quad (11.88)$$

This is exactly what we got from the first Born approximation (Equation 10.79).

Problem 11.17 In the photoelectric effect, light can ionize an atom if its energy ($\hbar\omega$) exceeds the binding energy of the electron. Consider the photoelectric effect for the ground state of hydrogen, where the electron is kicked out with momentum $\hbar\mathbf{k}$. The initial state of the electron is $\psi_0(r)$ (Equation 4.80) and its final state is²⁹

$$\psi_f = \frac{1}{\sqrt{l^3}} e^{i\mathbf{k} \cdot \mathbf{r}},$$

as in Example 11.2.

(a) For light polarized along the z direction, use Fermi's Golden Rule to compute the rate at which electrons are ejected into the solid angle $d\Omega$ in the dipole approximation.³⁰

$$\left[\text{Answer : } R_{i \rightarrow d\Omega} = 256 \frac{\epsilon_0 E_0^2 a^3}{\hbar} \frac{(ka)^3}{[1 + (ka)^2]^6} \cos^2 \theta d\Omega. \right]$$

²⁸ See footnote 24.

²⁹ This is an approximation; we really should be using a scattering state of hydrogen. For an extended discussion of the photoelectric effect, including comparison to experiment and the validity of this approximation, see W. Heitler, *The Quantum Theory of Radiation*, 3rd edn, Oxford University Press, London (1954), Section 21.

³⁰ The result here is too large by a factor of four; correcting this requires a more careful derivation of the matrix element for radiative transitions (see Problem 11.30). Only the overall factor is affected though; the more interesting features (the dependence on k and θ) are correct.

Hint: To evaluate the matrix element, use the following trick. Write

$$z e^{i \mathbf{k} \cdot \mathbf{r}} = -i \frac{d}{dk_z} e^{i \mathbf{k} \cdot \mathbf{r}},$$

pull d/dk_z outside the integral, and what remains is straightforward to compute.

(b) The **photoelectric cross section** is defined as

$$\sigma(k) = \frac{R_{i \rightarrow \text{all}} \hbar \omega}{\frac{1}{2} \epsilon_0 E_0^2 c}$$

where the quantity in the numerator is the rate at which *energy* is absorbed ($\hbar \omega = \frac{\hbar^2 k^2}{2m} - E_1$ per photoelectron) and the quantity in the denominator is the intensity of the incident light. Integrate your result from (a) over all angles to obtain $R_{i \rightarrow \text{all}}$, and compute the photoelectric cross section.

(c) Obtain a numerical value for the photoelectric cross section for ultraviolet light of wavelength 220 Å (n.b. this is the wavelength of the incident light, not the scattered electron). Express your answer in mega-barns ($\text{Mb} = 10^{-22} \text{ m}^2$).

11.5 THE ADIABATIC APPROXIMATION

11.5.1 Adiabatic Processes

Imagine a perfect pendulum, with no friction or air resistance, oscillating back and forth in a vertical plane. If you grab the support and shake it in a jerky manner the bob will swing around chaotically. But if you *very gently* move the support (Figure 11.12), the pendulum will continue to swing in a nice smooth way, in the same plane (or one parallel to it), with the same amplitude. This *gradual change of the external conditions* defines an **adiabatic** process. Notice that there are two characteristic times involved: T_i , the “internal” time, representing the motion of the system itself (in this case the period of the pendulum’s oscillations), and T_e , the “external” time, over which the parameters of the system change appreciably (if the pendulum were mounted on a rotating platform, for example, T_e would be the period of the *platform’s* motion). An adiabatic process is one for which $T_e \gg T_i$ (the pendulum executes many oscillations before the platform has moved appreciably).³¹

What if I took this pendulum up to the North Pole, and set it swinging—say, in the direction of Portland (Figure 11.13). For the moment, pretend the earth is not rotating. Very gently (that is, *adiabatically*), I carry it down the longitude line passing through Portland, to the equator. At this point it is swinging north-south. Now I carry it (still swinging north-south) part way around the equator. And finally, I take it back up to the North Pole, along the new longitude line. The pendulum will no longer be swinging in the same plane as it was when I set out—indeed, the new plane makes an angle Θ with the old one, where Θ is the angle between the southbound and the northbound longitude lines. More generally, if you transport the pendulum around a closed loop on the surface of the earth, the angular deviation (between the initial plane of the swing and the final plane) is equal to the solid angle subtended by the path with respect to the center of the sphere, as you can prove for yourself if you are interested.

Incidentally, the **Foucault pendulum** is an example of precisely this sort of adiabatic transport around a closed loop on a sphere—only this time instead of *me* carrying the pendulum

³¹ For an interesting discussion of classical adiabatic processes, see Frank S. Crawford, *Am. J. Phys.* **58**, 337 (1990).

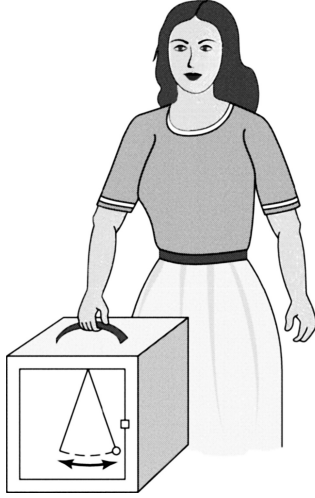


Figure 11.12: Adiabatic motion: If the case is transported very gradually, the pendulum inside keeps swinging with the same amplitude, in a plane parallel to the original one.

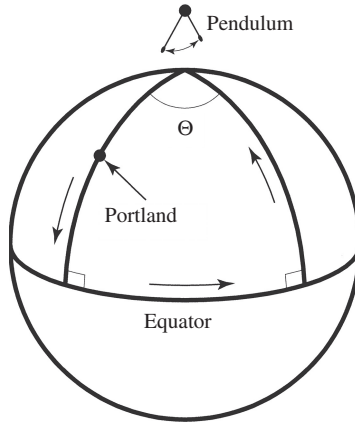


Figure 11.13: Itinerary for adiabatic transport of a pendulum on the surface of the earth.

around, I let the *rotation of the earth* do the job. The solid angle subtended by a latitude line θ_0 (Figure 11.14) is

$$\Omega = \int \sin \theta \, d\theta \, d\phi = 2\pi (-\cos \theta) \Big|_0^{\theta_0} = 2\pi (1 - \cos \theta_0). \quad (11.89)$$

Relative to the earth (which has meanwhile turned through an angle of 2π), the daily precession of the Foucault pendulum is $2\pi \cos \theta_0$ —a result that is ordinarily obtained by appeal to Coriolis forces in the rotating reference frame,³² but is seen in this context to admit a purely *geometrical* interpretation.

³² See, for example, Jerry B. Marion and Stephen T. Thornton, *Classical Dynamics of Particles and Systems*, 4th edn, Saunders, Fort Worth, TX (1995), Example 10.5. Geographers measure latitude (λ) up from the equator, rather than down from the pole, so $\cos \theta_0 = \sin \lambda$.

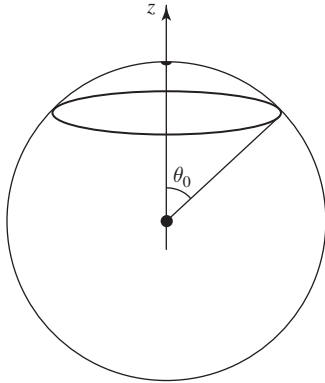


Figure 11.14: Path of a Foucault pendulum, in the course of one day.

The basic strategy for analyzing an adiabatic process is first to solve the problem with the external parameters held *constant*, and only at the *end* of the calculation allow them to vary (slowly) with time. For example, the classical period of a pendulum of (fixed) length L is $2\pi\sqrt{L/g}$; if the length is now gradually *changing*, the period will be $2\pi\sqrt{L(t)/g}$. A more subtle subtle example occurred in our discussion of the hydrogen molecule ion (Section 8.3). We began by assuming that the nuclei were *at rest*, a fixed distance R apart, and we solved for the motion of the electron. Once we had found the ground state energy of the system as a function of R , we located the equilibrium separation and from the curvature of the graph we obtained the frequency of vibration of the nuclei (Problem 8.11). In molecular physics this technique (beginning with nuclei at rest, calculating electronic wave functions, and using these to obtain information about the positions and—relatively sluggish—motion of the nuclei) is known as the **Born–Oppenheimer approximation**.

11.5.2 The Adiabatic Theorem

In quantum mechanics, the essential content of the **adiabatic approximation** can be cast in the form of a theorem. Suppose the Hamiltonian changes *gradually* from some initial form $\hat{H}(0)$ to some final form $\hat{H}(T)$. The **adiabatic theorem**³³ states that if the particle was initially in the n th eigenstate of $\hat{H}(0)$, it will be carried (under the Schrödinger equation) into the n th eigenstate of $\hat{H}(T)$. (I assume that the spectrum is discrete and nondegenerate throughout the transition, so there is no ambiguity about the ordering of the states; these conditions can be relaxed, given a suitable procedure for “tracking” the eigenfunctions, but I’m not going to pursue that here.)

Example 11.3

Suppose we prepare a particle in the ground state of the infinite square well (Figure 11.15(a)):

$$\psi^i(x) = \sqrt{\frac{2}{a}} \sin\left(\frac{\pi}{a}x\right). \quad (11.90)$$

³³ The adiabatic theorem, which is usually attributed to Ehrenfest, is simple to state, and it *sounds* plausible, but it is not easy to prove. The argument will be found in earlier editions of this book, Section 10.1.2.

If we now gradually move the right wall out to $2a$, the adiabatic theorem says that the particle will end up in the ground state of the expanded well (Figure 11.15(b)):

$$\psi^f(x) = \sqrt{\frac{1}{a}} \sin\left(\frac{\pi}{2a}x\right), \quad (11.91)$$

(apart from a phase factor, which we'll discuss in a moment). Notice that we're not talking about a *small* change in the Hamiltonian (as in perturbation theory)—this one is *huge*. All we require is that it happen *slowly*.

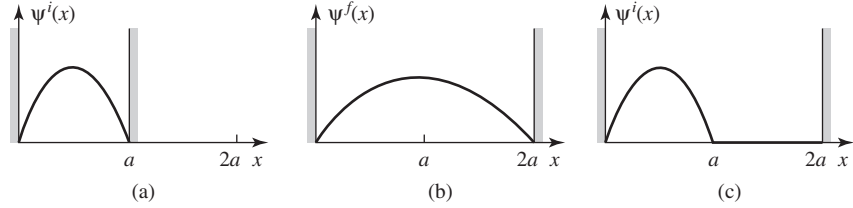


Figure 11.15: (a) Particle starts out in the ground state of the infinite square well. (b) If the wall moves *slowly*, the particle remains in the ground state. (c) If the wall moves *rapidly*, the particle is left (momentarily) in its initial state.

Energy is *not* conserved here—of *course* not: Whoever is moving the wall is extracting energy from the system, just like the piston on a slowly expanding cylinder of gas. By contrast, if the well expands *suddenly*, the resulting state is still $\psi^i(x)$ (Figure 11.15c), which is a complicated linear combination of eigenstates of the new Hamiltonian (Problem 11.18). In this case energy *is* conserved (at least, its *expectation value* is); as in the *free* expansion of a gas (into a vacuum) when the barrier is suddenly removed; no work is done.

According to the adiabatic theorem, a system that starts out in the n th eigenstate of the initial Hamiltonian ($\hat{H}(0)$) will evolve as the n th eigenstate of the instantaneous Hamiltonian ($\hat{H}(t)$), as the Hamiltonian gradually changes. However, this doesn't tell us what happens to the *phase* of the wave function. For a *constant* Hamiltonian it would pick up the standard “wobble factor”

$$e^{-iE_nt/\hbar},$$

but the eigenvalue E_n may now itself be a function of time, so the wobble factor naturally generalizes to

$$e^{i\theta_n(t)}, \quad \text{where} \quad \theta_n(t) \equiv -\frac{1}{\hbar} \int_0^t E_n(t') dt'. \quad (11.92)$$

This is called the **dynamic phase**. But it may not be the end of the story; for all we know there may be an additional phase factor, $\gamma_n(t)$, the so-called **geometric phase**. In the adiabatic limit, then, the wave function at time t takes the form³⁴

$$\Psi_n(t) = e^{i\theta_n(t)} e^{i\gamma_n(t)} \psi_n(t), \quad (11.93)$$

³⁴ I'm suppressing the dependence on other variables; only the time dependence is at issue here.

where $\psi_n(t)$ is the n th eigenstate of the instantaneous Hamiltonian,

$$\hat{H}(t)\psi_n(t) = E_n(t)\psi_n(t). \quad (11.94)$$

Equation 11.93 is the formal statement of the adiabatic theorem.

Of course, the phase of $\psi_n(t)$ is itself arbitrary (it's still an eigenfunction, with the same eigenvalue, whatever phase you choose), so the geometric phase itself carries no physical significance. But what if we carry the system around a *closed cycle* (like the pendulum we hauled down to the equator, around, and back to the north pole), so that the Hamiltonian at the end is identical to the Hamiltonian at the beginning? Then the *net* phase change is a measurable quantity. The dynamic phase depends on the elapsed time, but the geometric phase, around an adiabatic closed cycle, depends only on the path taken.³⁵ It is called **Berry's phase**.³⁶

$$\gamma_B \equiv \gamma(T) - \gamma(0). \quad (11.95)$$

Example 11.4

Imagine an electron (charge $-e$, mass m) at rest at the origin, in the presence of a magnetic field whose *magnitude* (B_0) is constant, but whose *direction* sweeps out a cone, of opening angle α , at constant angular velocity ω (Figure 11.16):

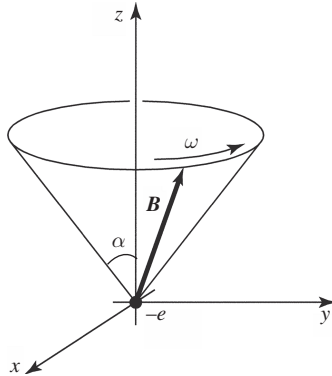


Figure 11.16: The magnetic field sweeps around in a cone, at angular velocity ω (Equation 11.96).

$$\mathbf{B}(t) = B_0 [\sin \alpha \cos(\omega t) \hat{i} + \sin \alpha \sin(\omega t) \hat{j} + \cos \alpha \hat{k}]. \quad (11.96)$$

The Hamiltonian (Equation 4.158) is

$$\begin{aligned} \hat{H}(t) &= \frac{e}{m} \mathbf{B} \cdot \mathbf{S} = \frac{e\hbar B_0}{2m} [\sin \alpha \cos(\omega t) \sigma_x + \sin \alpha \sin(\omega t) \sigma_y + \cos \alpha \sigma_z] \\ &= \frac{\hbar \omega_1}{2} \begin{pmatrix} \cos \alpha & e^{-i\omega t} \sin \alpha \\ e^{i\omega t} \sin \alpha & -\cos \alpha \end{pmatrix}, \end{aligned} \quad (11.97)$$

³⁵ As Michael Berry puts it, the dynamic phase answers the question “How long did your trip take?” and the geometric phase, “Where have you been?”

³⁶ For more on this subject see Alfred Shapere and Frank Wilczek, eds., *Geometric Phases in Physics*, World Scientific, Singapore (1989); Andrei Bernevig and Taylor Hughes, *Topological Insulators and Topological Superconductors*, Princeton University Press, Princeton, NJ (2013), Chapter 2.

where

$$\omega_1 \equiv \frac{eB_0}{m}. \quad (11.98)$$

The normalized eigenspinors of $\hat{H}(t)$ are

$$\chi_+(t) = \begin{pmatrix} \cos(\alpha/2) \\ e^{i\omega t} \sin(\alpha/2) \end{pmatrix}, \quad (11.99)$$

and

$$\chi_-(t) = \begin{pmatrix} e^{-i\omega t} \sin(\alpha/2) \\ -\cos(\alpha/2) \end{pmatrix}; \quad (11.100)$$

they represent spin up and spin down, respectively, *along the instantaneous direction of $\mathbf{B}(t)$* (see Problem 4.33). The corresponding eigenvalues are

$$E_{\pm} = \pm \frac{\hbar\omega_1}{2}. \quad (11.101)$$

Suppose the electron starts out with spin up, along $\mathbf{B}(0)$:

$$\chi(0) = \begin{pmatrix} \cos(\alpha/2) \\ \sin(\alpha/2) \end{pmatrix}. \quad (11.102)$$

The exact solution to the time-dependent Schrödinger equation is (Problem 11.20):

$$\chi(t) = \begin{pmatrix} \left[\cos(\lambda t/2) - i \frac{(\omega_1 - \omega)}{\lambda} \sin(\lambda t/2) \right] \cos(\alpha/2) e^{-i\omega t/2} \\ \left[\cos(\lambda t/2) - i \frac{(\omega_1 + \omega)}{\lambda} \sin(\lambda t/2) \right] \sin(\alpha/2) e^{+i\omega t/2} \end{pmatrix}, \quad (11.103)$$

where

$$\lambda \equiv \sqrt{\omega^2 + \omega_1^2 - 2\omega\omega_1 \cos \alpha}, \quad (11.104)$$

or, expressing it as a linear combination of χ_+ and χ_- :

$$\begin{aligned} \chi(t) = & \left[\cos\left(\frac{\lambda t}{2}\right) - i \frac{(\omega_1 - \omega \cos \alpha)}{\lambda} \sin\left(\frac{\lambda t}{2}\right) \right] e^{-i\omega t/2} \chi_+(t) \\ & + i \left[\frac{\omega}{\lambda} \sin \alpha \sin\left(\frac{\lambda t}{2}\right) \right] e^{+i\omega t/2} \chi_-(t). \end{aligned} \quad (11.105)$$

Evidently the (exact) probability of a transition to spin down (along the current direction of $\mathbf{B}$) is

$$|\langle \chi(t) | \chi_-(t) \rangle|^2 = \left[\frac{\omega}{\lambda} \sin \alpha \sin\left(\frac{\lambda t}{2}\right) \right]^2. \quad (11.106)$$

The adiabatic theorem says that this transition probability should vanish in the limit $T_e \gg T_i$, where T_e is the characteristic time for changes in the Hamiltonian (in this case, $1/\omega$) and T_i is the characteristic time for changes in the wave function (in this case, $\hbar/(E_+ - E_-) = 1/\omega_1$). Thus the adiabatic approximation means $\omega \ll \omega_1$: the field rotates slowly, in comparison with the phase of the (unperturbed) wave functions. In the adiabatic regime $\lambda \approx \omega_1$ (Equation 11.104), and therefore

$$|\langle \chi(t) | \chi_-(t) \rangle|^2 \approx \left[\frac{\omega}{\omega_1} \sin \alpha \sin\left(\frac{\lambda t}{2}\right) \right]^2 \rightarrow 0, \quad (11.107)$$

as advertised. The magnetic field leads the electron around by its nose, with the spin always pointing in the direction of $\mathbf{B}$. By contrast, if $\omega \gg \omega_1$ then $\lambda \approx \omega$, and the system bounces back and forth between spin up and spin down (Figure 11.17).

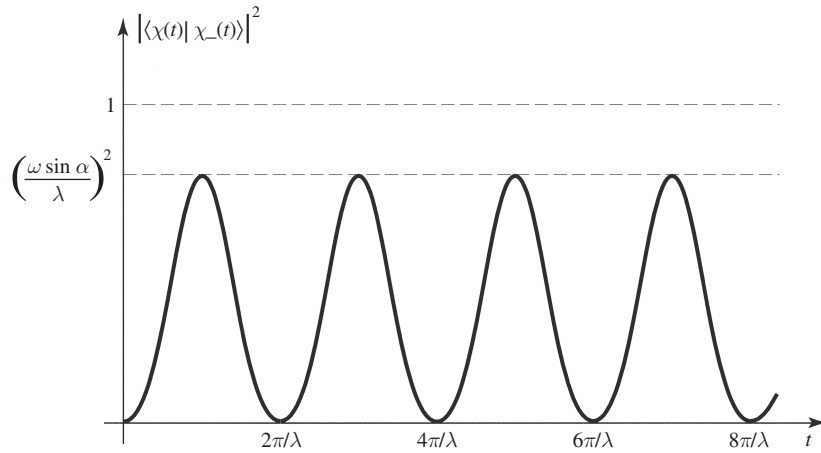


Figure 11.17: Plot of the transition probability, Equation 11.106, in the *non*-adiabatic regime ($\omega \gg \omega_1$).

* **Problem 11.18** A particle of mass m is in the ground state of the infinite square well (Equation 2.22). Suddenly the well expands to twice its original size—the right wall moving from a to $2a$ —leaving the wave function (momentarily) undisturbed. The energy of the particle is now measured.

- What is the most probable result? What is the probability of getting that result?
- What is the *next* most probable result, and what is its probability? Suppose your measurement returned this value; what would you conclude about conservation of energy?
- What is the *expectation value* of the energy? *Hint:* If you find yourself confronted with an infinite series, try another method.

Problem 11.19 A particle is in the ground state of the harmonic oscillator with classical frequency ω , when suddenly the spring constant quadruples, so $\omega' = 2\omega$, without initially changing the wave function (of course, Ψ will now *evolve* differently, because the Hamiltonian has changed). What is the probability that a measurement of the energy would still return the value $\hbar\omega/2$? What is the probability of getting $\hbar\omega$? *Answer:* 0.943.

** **Problem 11.20** Check that Equation 11.103 satisfies the time-dependent Schrödinger equation for the Hamiltonian in Equation 11.97. Also confirm Equation 11.105, and show that the sum of the squares of the coefficients is 1, as required for normalization.

- * **Problem 11.21** Find Berry's phase for one cycle of the process in Example 11.4. *Hint:* Use Equation 11.105 to determine the *total* phase change, and subtract off the dynamical part. You'll need to expand λ (Equation 11.104) to first order in ω/ω_1 .

Problem 11.22 The delta function well (Equation 2.117) supports a single bound state (Equation 2.132). Calculate the geometric phase change when α gradually increases from α_1 to α_2 . If the increase occurs at a constant rate ($d\alpha/dt = c$), what is the dynamic phase change for this process?³⁷

FURTHER PROBLEMS ON CHAPTER 11

- *** **Problem 11.23** In Problem 11.1 you showed that the solution to

$$\frac{df}{dt} = k(t) f(t)$$

(where $k(t)$ is a function of t) is

$$f(t) = e^{K(t)} f(0), \quad \text{where} \quad K(t) \equiv \int_0^t k(t') dt'.$$

This suggests that the solution to the Schrödinger equation (11.1) might be

$$\Psi(t) = e^{\hat{G}(t)} \Psi(0), \quad \text{where} \quad \hat{G}(t) \equiv -\frac{i}{\hbar} \int_0^t \hat{H}(t') dt'. \quad (11.108)$$

It doesn't work, because $\hat{H}(t)$ is an *operator*, not a function, and $\hat{H}(t_1)$ does not (in general) commute with $\hat{H}(t_2)$.

- (a) Try calculating $i\hbar \partial\Psi/\partial t$, using Equation 11.108. *Note:* as always, the exponentiated operator is to be interpreted as a power series:

$$e^{\hat{G}} \equiv 1 + \hat{G} + \frac{1}{2} \hat{G} \hat{G} + \frac{1}{3!} \hat{G} \hat{G} \hat{G} + \dots$$

Show that if $[\hat{G}, \hat{H}] = 0$, then Ψ satisfies the Schrödinger equation.

- (b) Check that the correct solution in the general case ($[\hat{G}, \hat{H}] \neq 0$) is

$$\begin{aligned} \Psi(t) = & \left\{ 1 + \left(-\frac{i}{\hbar}\right) \int_0^t \hat{H}(t_1) dt_1 + \left(-\frac{i}{\hbar}\right)^2 \int_0^t \hat{H}(t_1) \left[\int_0^{t_1} \hat{H}(t_2) dt_2 \right] dt_1 \right. \\ & \left. + \left(-\frac{i}{\hbar}\right)^3 \int_0^t \hat{H}(t_1) \left[\int_0^{t_1} \hat{H}(t_2) \left(\int_0^{t_2} \hat{H}(t_3) dt_3 \right) dt_2 \right] dt_1 + \dots \right\} \Psi(0). \end{aligned} \quad (11.109)$$

³⁷ If $\psi_n(t)$ is *real*, the geometric phase vanishes. You might try to beat the rap by tacking an unnecessary (but perfectly legal) phase factor onto the eigenfunctions: $\psi'_n(t) \equiv e^{i\phi_n} \psi_n(t)$, where ϕ_n is an arbitrary (real) function. Try it. You'll get a nonzero geometric phase, all right, but note what happens when you put it back into Equation 11.93. And for a *closed* loop it gives *zero*.

UGLY! Notice that the operators in each term are “time-ordered,” in the sense that the *latest* $\hat{H}$ appears at the far left, followed by the next latest, and so on ($t \geq t_1 \geq t_2 \geq t_3 \cdots$). Dyson introduced the **time-ordered product** of two operators:

$$\mathbf{T}[\hat{H}(t_i)\hat{H}(t_j)] \equiv \begin{cases} \hat{H}(t_i)\hat{H}(t_j), & t_i \geq t_j \\ \hat{H}(t_j)\hat{H}(t_i), & t_j \geq t_i \end{cases} \quad (11.110)$$

or, more generally,

$$\mathbf{T}[\hat{H}(t_1)\hat{H}(t_2)\cdots\hat{H}(t_n)] \equiv \hat{H}(t_{j_1})\hat{H}(t_{j_2})\cdots\hat{H}(t_{j_n}), \quad (11.111)$$

where $t_{j_1} \geq t_{j_2} \geq \cdots \geq t_{j_n}$.

(c) Show that

$$\mathbf{T}[\hat{G}\hat{G}] = -\frac{2}{\hbar^2} \int_0^t \hat{H}(t_1) \left[\int_0^{t_1} \hat{H}(t_2) dt_2 \right] dt_1,$$

and generalize to higher powers of $\hat{G}$. In place of $\hat{G}^n$, in equation 11.108, we really want $\mathbf{T}[\hat{G}^n]$:

$$\Psi(t) = \mathbf{T}\left[e^{-\frac{i}{\hbar} \int_0^t \hat{H}(t') dt'}\right] \Psi(0). \quad (11.112)$$

This is **Dyson’s formula**; it’s a compact way of writing Equation 11.109, the formal solution to Schrödinger’s equation. Dyson’s formula plays a fundamental role in quantum field theory.³⁸

**

Problem 11.24 In this problem we develop time-dependent perturbation theory for a multi-level system, starting with the generalization of Equations 11.5 and 11.6:

$$\hat{H}_0 \psi_n = E_n \psi_n, \quad \langle \psi_n | \psi_m \rangle = \delta_{nm}. \quad (11.113)$$

At time $t = 0$ we turn on a perturbation $H'(t)$, so that the total Hamiltonian is

$$\hat{H} = \hat{H}_0 + \hat{H}'(t). \quad (11.114)$$

³⁸ The **interaction picture** is intermediate between the Heisenberg and Schrödinger pictures (see Section 6.8.1). In the interaction picture, the wave function satisfies the “Schrödinger equation”

$$i\hbar \frac{d}{dt} |\Psi_I(t)\rangle = \hat{H}'_I(t) |\Psi_I(t)\rangle,$$

where the interaction- and Schrödinger-picture operators are related by

$$\hat{H}'_I(t) = e^{i\hat{H}_0 t/\hbar} \hat{H}'(t) e^{-i\hat{H}_0 t/\hbar}$$

and the wave functions satisfy

$$|\Psi_I(t)\rangle = e^{i\hat{H}_0 t/\hbar} |\Psi(t)\rangle.$$

If you apply the Dyson series to the Schrödinger equation in the interaction picture, you end up with precisely the perturbation series derived in Section 11.1.2. For more details see Ramamurti Shankar, *Principles of Quantum Mechanics*, 2nd edn, Springer, New York (1994), Section 18.3.

(a) Generalize Equation 11.10 to read

$$\Psi(t) = \sum_n c_n(t) \psi_n e^{-iE_n t/\hbar}, \quad (11.115)$$

and show that

$$\dot{c}_m = -\frac{i}{\hbar} \sum_n c_n H'_{mn} e^{i(E_m - E_n)t/\hbar}, \quad (11.116)$$

where

$$H'_{mn} \equiv \langle \psi_m | \hat{H}' | \psi_n \rangle. \quad (11.117)$$

(b) If the system starts out in the state ψ_N , show that (in first-order perturbation theory)

$$c_N(t) \approx 1 - \frac{i}{\hbar} \int_0^t H'_{NN}(t') dt'. \quad (11.118)$$

and

$$c_m(t) \approx -\frac{i}{\hbar} \int_0^t H'_{mN}(t') e^{i(E_m - E_N)t'/\hbar} dt', \quad (m \neq N). \quad (11.119)$$

(c) For example, suppose $\hat{H}'$ is *constant* (except that it was turned on at $t = 0$, and switched off again at some later time T). Find the probability of transition from state N to state M ($M \neq N$), as a function of T . *Answer:*

$$4 |H'_{MN}|^2 \frac{\sin^2[(E_N - E_M) T/2\hbar]}{(E_N - E_M)^2}. \quad (11.120)$$

(d) Now suppose $\hat{H}'$ is a sinusoidal function of time: $\hat{H}' = V \cos(\omega t)$. Making the usual assumptions, show that transitions occur only to states with energy $E_M = E_N \pm \hbar\omega$, and the transition probability is

$$P_{N \rightarrow M} = |V_{MN}|^2 \frac{\sin^2[(E_N - E_M \pm \hbar\omega) T/2\hbar]}{(E_N - E_M \pm \hbar\omega)^2}. \quad (11.121)$$

(e) Suppose a multi-level system is immersed in incoherent electromagnetic radiation. Using Section 11.2.3 as a guide, show that the transition rate for stimulated emission is given by the same formula (Equation 11.54) as for a two-level system.

Problem 11.25 For the examples in Problem 11.24 (c) and (d), calculate $c_m(t)$, to first order. Check the normalization condition:

$$\sum_m |c_m(t)|^2 = 1, \quad (11.122)$$

and comment on any discrepancy. Suppose you wanted to calculate the probability of *remaining* in the original state ψ_N ; would you do better to use $|c_N(t)|^2$, or $1 - \sum_{m \neq N} |c_m(t)|^2$?

Problem 11.26 A particle starts out (at time $t = 0$) in the N th state of the infinite square well. Now the “floor” of the well rises temporarily (maybe water leaks in, and then drains out again), so that the potential inside is uniform but time dependent: $V_0(t)$, with $V_0(0) = V_0(T) = 0$.

- (a) Solve for the *exact* $c_m(t)$, using Equation 11.116, and show that the wave function changes *phase*, but no transitions occur. Find the phase change, $\phi(T)$, in terms of the function $V_0(t)$.
- (b) Analyze the same problem in first-order perturbation theory, and compare your answers.

Comment: The same result holds *whenever* the perturbation simply adds a constant (constant in x , that is, not in t) to the potential; it has nothing to do with the infinite square well, as such. Compare Problem 1.8.

- * **Problem 11.27** A particle of mass m is initially in the ground state of the (one-dimensional) infinite square well. At time $t = 0$ a “brick” is dropped into the well, so that the potential becomes

$$V(x) = \begin{cases} V_0, & 0 \leq x \leq a/2, \\ 0, & a/2 < x \leq a, \\ \infty, & \text{otherwise,} \end{cases}$$

where $V_0 \ll E_1$. After a time T , the brick is removed, and the energy of the particle is measured. Find the probability (in first-order perturbation theory) that the result is now E_2 .

Problem 11.28 We have encountered stimulated emission, (stimulated) absorption, and spontaneous emission. How come there is no such thing as spontaneous *absorption*?

- *** **Problem 11.29 Magnetic resonance.** A spin-1/2 particle with gyromagnetic ratio γ , at rest in a static magnetic field $B_0 \hat{k}$, precesses at the Larmor frequency $\omega_0 = \gamma B_0$ (Example 4.3). Now we turn on a small transverse radiofrequency (rf) field, $B_{\text{rf}} [\cos(\omega t) \hat{i} - \sin(\omega t) \hat{j}]$, so that the total field is

$$\mathbf{B} = B_{\text{rf}} \cos(\omega t) \hat{i} - B_{\text{rf}} \sin(\omega t) \hat{j} + B_0 \hat{k}. \quad (11.123)$$

(a) Construct the 2×2 Hamiltonian matrix (Equation 4.158) for this system.

(b) If $\chi(t) = \begin{pmatrix} a(t) \\ b(t) \end{pmatrix}$ is the spin state at time t , show that

$$\dot{a} = \frac{i}{2} (\Omega e^{i\omega t} b + \omega_0 a); \quad \dot{b} = \frac{i}{2} (\Omega e^{-i\omega t} a - \omega_0 b), \quad (11.124)$$

where $\Omega \equiv \gamma B_{\text{rf}}$ is related to the strength of the rf field.

(c) Check that the general solution for $a(t)$ and $b(t)$, in terms of their initial values a_0 and b_0 , is

$$a(t) = \left\{ a_0 \cos(\omega' t/2) + \frac{i}{\omega'} [a_0 (\omega_0 - \omega) + b_0 \Omega] \sin(\omega' t/2) \right\} e^{i\omega t/2}$$

$$b(t) = \left\{ b_0 \cos(\omega' t/2) + \frac{i}{\omega'} [b_0 (\omega - \omega_0) + a_0 \Omega] \sin(\omega' t/2) \right\} e^{-i\omega t/2}$$

where

$$\omega' \equiv \sqrt{(\omega - \omega_0)^2 + \Omega^2}. \quad (11.125)$$

- (d) If the particle starts out with spin up (i.e. $a_0 = 1$, $b_0 = 0$), find the probability of a transition to spin down, as a function of time. *Answer:* $P(t) = \{\Omega^2 / [(\omega - \omega_0)^2 + \Omega^2]\} \sin^2(\omega' t/2)$.
- (e) Sketch the **resonance curve**,

$$P(\omega) = \frac{\Omega^2}{(\omega - \omega_0)^2 + \Omega^2}, \quad (11.126)$$

as a function of the driving frequency ω (for fixed ω_0 and Ω). Note that the maximum occurs at $\omega = \omega_0$. Find the “full width at half maximum,” $\Delta\omega$.

- (f) Since $\omega_0 = \gamma B_0$, we can use the experimentally observed resonance to determine the magnetic dipole moment of the particle. In a **nuclear magnetic resonance** (nmr) experiment the g -factor of the proton is to be measured, using a static field of 10,000 gauss and an rf field of amplitude 0.01 gauss. What will the resonant frequency be? (See Section 7.5 for the magnetic moment of the proton.) Find the width of the resonance curve. (Give your answers in Hz.)

**

Problem 11.30 In this problem we will recover the results Section 11.2.1 directly from the Hamiltonian for a charged particle in an electromagnetic field (Equation 4.188). An electromagnetic wave can be described by the potentials

$$\mathbf{A} = \frac{\mathbf{E}_0}{\omega} \sin(\mathbf{k} \cdot \mathbf{r} - \omega t), \quad \phi = 0$$

where in order to satisfy Maxwell's equations, the wave must be transverse ($\mathbf{E}_0 \cdot \mathbf{k} = 0$) and of course travel at the speed of light ($\omega = c |\mathbf{k}|$).

- (a) Find the electric and magnetic fields for this plane wave.
- (b) The Hamiltonian may be written as $H^0 + H'$ where H^0 is the Hamiltonian in the absence of the electromagnetic wave and H' is the perturbation. Show that the perturbation is given by

$$\hat{H}'(t) = \frac{e}{2im\omega} e^{i\mathbf{k} \cdot \mathbf{r}} \mathbf{E}_0 \cdot \hat{\mathbf{p}} e^{-i\omega t} - \frac{e}{2im\omega} e^{-i\mathbf{k} \cdot \mathbf{r}} \mathbf{E}_0 \cdot \hat{\mathbf{p}} e^{i\omega t}, \quad (11.127)$$

plus a term proportional to E_0^2 that we will ignore. *Note:* the first term corresponds to absorption and the second to emission.

- (c) In the dipole approximation we set $e^{i\mathbf{k} \cdot \mathbf{r}} \approx 1$. With the electromagnetic wave polarized along the z direction, show that the matrix element for absorption is then

$$V_{ba} = -\frac{\omega_0}{\omega} \wp E_0.$$

Compare Equation 11.41. They're not *exactly* the same; would the difference effect our calculations in Section 11.2.3 or 11.3? Why or why not? *Hint:* To turn the matrix element of $\mathbf{p}$ into a matrix element of $\mathbf{r}$, you need to prove the following identity: $i m [\hat{H}^0, \hat{\mathbf{r}}] = \hbar \hat{\mathbf{p}}$.

Problem 11.31 In Equation 11.38 I assumed that the atom is so small (in comparison to the wavelength of the light) that spatial variations in the field can be ignored. The *true* electric field would be

$$\mathbf{E}(\mathbf{r}, t) = \mathbf{E}_0 \cos(\mathbf{k} \cdot \mathbf{r} - \omega t). \quad (11.128)$$

If the atom is centered at the origin, then $\mathbf{k} \cdot \mathbf{r} \ll 1$ over the relevant volume ($|\mathbf{k}| = 2\pi/\lambda$, so $\mathbf{k} \cdot \mathbf{r} \sim r/\lambda \ll 1$), and that's why we could afford to drop this term. Suppose we keep the first-order correction:

$$\mathbf{E}(\mathbf{r}, t) = \mathbf{E}_0 [\cos(\omega t) + (\mathbf{k} \cdot \mathbf{r}) \sin(\omega t)]. \quad (11.129)$$

The first term gives rise to the **allowed (electric dipole)** transitions we considered in the text; the second leads to so-called **forbidden (magnetic dipole and electric quadrupole)** transitions (higher powers of $\mathbf{k} \cdot \mathbf{r}$ lead to even *more* “forbidden” transitions, associated with higher multipole moments).³⁹

(a) Obtain the spontaneous emission rate for forbidden transitions (don't bother to average over polarization and propagation directions, though this should really be done to complete the calculation). *Answer:*

$$R_{b \rightarrow a} = \frac{q^2 \omega^5}{\pi \epsilon_0 \hbar c^5} \left| \left\langle a \left| (\hat{n} \cdot \mathbf{r}) (\hat{k} \cdot \mathbf{r}) \right| b \right\rangle \right|^2. \quad (11.130)$$

(b) Show that for a one-dimensional oscillator the forbidden transitions go from level n to level $n - 2$, and the transition rate (suitably averaged over $\hat{n}$ and $\hat{k}$) is

$$R = \frac{\hbar q^2 \omega^3 n (n - 1)}{15 \pi \epsilon_0 m^2 c^5}. \quad (11.131)$$

(*Note:* Here ω is the frequency of the *photon*, not the oscillator.) Find the *ratio* of the “forbidden” rate to the “allowed” rate, and comment on the terminology.

(c) Show that the $2S \rightarrow 1S$ transition in hydrogen is not possible even by a “forbidden” transition. (As it turns out, this is true for all the higher multipoles as well; the dominant decay is in fact by two-photon emission, and the lifetime is about a tenth of a second.⁴⁰)

³⁹ For a systematic treatment (including the role of the magnetic field) see David Park, *Introduction to the Quantum Theory*, 3rd edn (McGraw-Hill, New York, 1992), Chapter 11.

⁴⁰ See Masataka Mizushima, *Quantum Mechanics of Atomic Spectra and Atomic Structure*, Benjamin, New York (1970), Section 5.6.

Problem 11.32 Show that the spontaneous emission rate (Equation 11.63) for a transition from n, ℓ to n', ℓ' in hydrogen is

$$\frac{e^2 \omega^3 I^2}{3\pi \epsilon_0 \hbar c^3} \times \begin{cases} \frac{\ell+1}{2\ell+1}, & \ell' = \ell + 1, \\ \frac{\ell}{2\ell+1} & \ell' = \ell - 1, \end{cases} \quad (11.132)$$

where

$$I \equiv \int_0^\infty r^3 R_{n\ell}(r) R_{n'\ell'}(r) dr. \quad (11.133)$$

(The atom starts out with a specific value of m , and it goes to *any* of the states m' consistent with the selection rules: $m' = m + 1, m$, or $m - 1$. Notice that the answer is independent of m .) *Hint*: First calculate all the nonzero matrix elements of x , y , and z between $|n\ell m\rangle$ and $|n'\ell'm'\rangle$ for the case $\ell' = \ell + 1$. From these, determine the quantity

$$|\langle n', \ell + 1, m + 1 | \mathbf{r} | n\ell m \rangle|^2 + |\langle n', \ell + 1, m | \mathbf{r} | n\ell m \rangle|^2 + |\langle n', \ell + 1, m - 1 | \mathbf{r} | n\ell m \rangle|^2.$$

Then do the same for $\ell' = \ell - 1$. You may find useful the following recursion formulas (which hold for $m \geq 0$):⁴¹

$$(2\ell + 1)x P_\ell^m(x) = (\ell + m) P_{\ell-1}^m(x) + (\ell - m + 1) P_{\ell+1}^m(x), \quad (11.134)$$

$$(2\ell + 1)\sqrt{1 - x^2} P_\ell^m(x) = P_{\ell+1}^{m+1}(x) - P_{\ell-1}^{m+1}(x), \quad (11.135)$$

and the orthogonality relation Equation 4.33.

Problem 11.33 The spontaneous emission rate for the 21-cm hyperfine line in hydrogen (Section 7.5) can be obtained from Equation 11.63, except that this is a *magnetic* dipole transition, not an *electric* one:⁴²

$$\mathfrak{P} \rightarrow \frac{1}{c} \mathbf{M} = \frac{1}{c} \langle 1 | (\boldsymbol{\mu}_e + \boldsymbol{\mu}_p) | 0 \rangle,$$

where

$$\boldsymbol{\mu}_e = -\frac{e}{m_e} \mathbf{S}_e, \quad \boldsymbol{\mu}_p = \frac{5.59 e}{2m_p} \mathbf{S}_p$$

are the magnetic moments of the electron and proton (Equation 7.89), and $|0\rangle, |1\rangle$ are the singlet and triplet configurations (Equations 4.175 and 4.176). Because $m_p \gg m_e$, the proton contribution is negligible, so

$$A = \frac{\omega_0^3 e^2}{3\pi \epsilon_0 \hbar c^5 m_e^2} |\langle 1 | \mathbf{S}_e | 0 \rangle|^2.$$

Work out $|\langle 1 | \mathbf{S}_e | 0 \rangle|^2$ (use whichever triplet state you like). Put in the actual numbers, to determine the transition rate and the lifetime of the triplet state. *Answer*: 1.1×10^7 years.

⁴¹ George B. Arfken and Hans J. Weber, *Mathematical Methods for Physicists*, 7th edn, Academic Press, San Diego (2013), p. 744.

⁴² Electric and magnetic dipole moments have different units—hence the factor of $1/c$ (which you can check by dimensional analysis).



Problem 11.34 A particle starts out in the ground state of the infinite square well (on the interval $0 \leq x \leq a$). Now a wall is slowly erected, slightly off-center:⁴³

$$V(x) = f(t)\delta\left(x - \frac{a}{2} - \epsilon\right),$$

where $f(t)$ rises gradually from 0 to ∞ . According to the adiabatic theorem, the particle will remain in the ground state of the evolving Hamiltonian.

- (a) Find (and sketch) the ground state at $t \rightarrow \infty$. *Hint:* This should be the ground state of the infinite square well with an impenetrable barrier at $a/2 + \epsilon$. Note that the particle is confined to the (slightly) larger left “half” of the well.
- (b) Find the (transcendental) equation for the ground state energy at time t . *Answer:*

$$z \sin z = T [\cos z - \cos(z\delta)],$$

where $z \equiv ka$, $T \equiv maf(t)/\hbar^2$, $\delta \equiv 2\epsilon/a$, and $k \equiv \sqrt{2mE}/\hbar$.

- (c) Setting $\delta = 0$, solve graphically for z , and show that the smallest z goes from π to 2π as T goes from 0 to ∞ . Explain this result.
- (d) Now set $\delta = 0.01$ and solve numerically for z , using $T = 0, 1, 5, 20, 100$, and 1000.
- (e) Find the probability P_r that the particle is in the right “half” of the well, as a function of z and δ . *Answer:* $P_r = 1/[1 + (I_+/I_-)]$, where $I_{\pm} \equiv [1 \pm \delta - (1/z) \sin(z(1 \pm \delta))] \sin^2[z(1 \mp \delta)/2]$. Evaluate this expression numerically for the T 's and δ in part (d). Comment on your results.
- (e) Plot the ground state wave function for those same values of T and δ . Note how it gets squeezed into the left half of the well, as the barrier grows.⁴⁴

Problem 11.35 The case of an infinite square well whose right wall expands at a *constant* velocity (v) can be solved *exactly*.⁴⁵ A complete set of solutions is

$$\Phi_n(x, t) \equiv \sqrt{\frac{2}{w}} \sin\left(\frac{n\pi}{w}x\right) e^{i(mvx^2 - 2E_n^i t)/2\hbar w}, \quad (11.136)$$

where $w(t) \equiv a + vt$ is the width of the well and $E_n^i \equiv n^2\pi^2\hbar^2/2ma^2$ is the n th allowed energy of the *original* well (width a). The general solution is a linear combination of the Φ 's:

$$\Psi(x, t) = \sum_{n=1}^{\infty} c_n \Phi_n(x, t), \quad (11.137)$$

whose coefficients c_n are *independent of t* .

- (a) Check that Equation 11.136 satisfies the time-dependent Schrödinger equation, with the appropriate boundary conditions.

⁴³ Julio Gea-Banacloche, *Am. J. Phys.* **70**, 307 (2002) uses a rectangular barrier; the delta-function version was suggested by M. Lakner and J. Peterelj, *Am. J. Phys.* **71**, 519 (2003).

⁴⁴ Gea-Banacloche (footnote 43) discusses the evolution of the wave function *without* using the adiabatic theorem.

⁴⁵ S. W. Doescher and M. H. Rice, *Am. J. Phys.* **37**, 1246 (1969).

- (b) Suppose a particle starts out ($t = 0$) in the ground state of the initial well:

$$\Psi(x, 0) = \sqrt{\frac{2}{a}} \sin\left(\frac{\pi}{a}x\right).$$

Show that the expansion coefficients can be written in the form

$$c_n = \frac{2}{\pi} \int_0^\pi e^{-i\alpha z^2} \sin(nz) \sin(z) dz, \quad (11.138)$$

where $\alpha \equiv mva/2\pi^2\hbar$ is a dimensionless measure of the speed with which the well expands. (Unfortunately, this integral cannot be evaluated in terms of elementary functions.)

- (c) Suppose we allow the well to expand to twice its original width, so the “external” time is given by $w(T_e) = 2a$. The “internal” time is the *period* of the time-dependent exponential factor in the (initial) ground state. Determine T_e and T_i , and show that the adiabatic regime corresponds to $\alpha \ll 1$, so that $\exp(-i\alpha z^2) \approx 1$ over the domain of integration. Use this to determine the expansion coefficients, c_n . Construct $\Psi(x, t)$, and confirm that it is consistent with the adiabatic theorem.
- (d) Show that the phase factor in $\Psi(x, t)$ can be written in the form

$$\theta(t) = -\frac{1}{\hbar} \int_0^t E_1(t') dt', \quad (11.139)$$

where $E_n(t) \equiv n^2\pi^2\hbar^2/2mw^2$ is the n th *instantaneous* eigenvalue, at time t . Comment on this result. What is the geometric phase? If the well now contracts back to its original size, what is Berry’s phase for the cycle?

Problem 11.36 The driven harmonic oscillator. Suppose the one-dimensional harmonic oscillator (mass m , frequency ω) is subjected to a driving force of the form $F(t) = m\omega^2 f(t)$, where $f(t)$ is some specified function. (I have factored out $m\omega^2$ for notational convenience; $f(t)$ has the dimensions of *length*.) The Hamiltonian is

$$H(t) = -\frac{\hbar^2}{2m} \frac{\partial^2}{\partial x^2} + \frac{1}{2}m\omega^2 x^2 - m\omega^2 x f(t). \quad (11.140)$$

Assume that the force was first turned on at time $t = 0$: $f(t) = 0$ for $t \leq 0$. This system can be solved exactly, both in classical mechanics and in quantum mechanics.⁴⁶

- (a) Determine the *classical* position of the oscillator, assuming it started from rest at the origin ($x_c(0) = \dot{x}_c(0) = 0$). *Answer:*

$$x_c(t) = \omega \int_0^t f(t') \sin[\omega(t - t')] dt'. \quad (11.141)$$

- (b) Show that the solution to the (time-dependent) Schrödinger equation for this oscillator, assuming it started out in the n th state of the *undriven* oscillator ($\Psi(x, 0) = \psi_n(x)$ where $\psi_n(x)$ is given by Equation 2.62), can be written as

⁴⁶ See Y. Nogami, *Am. J. Phys.* **59**, 64 (1991), and references therein.

$$\Psi(x, t) = \psi_n(x - x_c) e^{\frac{i}{\hbar} \left[-\left(n + \frac{1}{2}\right) \hbar \omega t + m \dot{x}_c \left(x - \frac{x_c}{2}\right) + \frac{m \omega^2}{2} \int_0^t f(t') x_c(t') dt' \right]}. \quad (11.142)$$

(c) Show that the eigenfunctions and eigenvalues of $H(t)$ are

$$\psi_n(x, t) = \psi_n(x - f); \quad E_n(t) = \left(n + \frac{1}{2}\right) \hbar \omega - \frac{1}{2} m \omega^2 f^2. \quad (11.143)$$

(d) Show that in the adiabatic approximation the classical position (Equation 11.141) reduces to $x_c(t) \approx f(t)$. State the precise criterion for adiabaticity, in this context, as a constraint on the time derivative of f . *Hint:* Write $\sin[\omega(t - t')]$ as $(1/\omega) (d/dt') \cos[\omega(t - t')]$ and use integration by parts.

(e) Confirm the adiabatic theorem for this example, by using the results in (c) and (d) to show that

$$\Psi(x, t) \approx \psi_n(x, t) e^{i\theta_n(t)} e^{i\gamma_n(t)}. \quad (11.144)$$

Check that the dynamic phase has the correct form (Equation 11.92). Is the geometric phase what you would expect?

Problem 11.37 Quantum Zeno Paradox.⁴⁷ Suppose a system starts out in an excited state ψ_b , which has a natural lifetime τ for transition to the ground state ψ_a . Ordinarily, for times substantially less than τ , the probability of a transition is proportional to t (Equation 11.49):

$$P_{b \rightarrow a} = \frac{t}{\tau}. \quad (11.145)$$

If we make a measurement after a time t , then, the probability that the system is still in the *upper* state is

$$P_b(t) = 1 - \frac{t}{\tau}. \quad (11.146)$$

Suppose we *do* find it to be in the upper state. In that case the wave function collapses back to ψ_b , and the process starts all over again. If we make a *second* measurement, at $2t$, the probability that the system is *still* in the upper state is

$$\left(1 - \frac{t}{\tau}\right)^2 \approx 1 - \frac{2t}{\tau}, \quad (11.147)$$

which is the same as it would have been had we never made the first measurement at t (as one would naively expect).

However, for *extremely* short times, the probability of a transition is *not* proportional to t , but rather to t^2 (Equation 11.46):⁴⁸

$$P_{b \rightarrow a} = \alpha t^2. \quad (11.148)$$

⁴⁷ This phenomenon doesn't have much to do with Zeno, but it *is* reminiscent of the old adage, "a watched pot never boils," so it is sometimes called the **watched pot effect**.

⁴⁸ In the argument leading to linear time dependence, we assumed that the function $\sin^2(\Omega t/2)/\Omega^2$ in Equation 11.46 was a sharp spike. However, the *width* of the "spike" is of order $\Delta\omega = 4\pi/t$, and for *extremely* short t this assumption fails, and the integral becomes $(t^2/4) \int \rho(\omega) d\omega$.

- (a) In this case what is the probability that the system is still in the upper state after the two measurements? What *would* it have been (after the same elapsed time) if we had never made the first measurement?
- (b) Suppose we examine the system at n regular (extremely short) intervals, from $t = 0$ out to $t = T$ (that is, we make measurements at $T/n, 2T/n, 3T/n, \dots, T$). What is the probability that the system is still in the upper state at time T ? What is its limit as $n \rightarrow \infty$? *Moral of the story:* Because of the collapse of the wave function at every measurement, a *continuously* observed system never decays at all!⁴⁹

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Problem 11.38 The numerical solution to the time-independent Schrödinger equation in Problem 2.61 can be extended to solve the time-dependent Schrödinger equation. When we discretize the variable x , we obtain the matrix equation

$$H \Psi = i \hbar \frac{d}{dt} \Psi. \quad (11.149)$$

The solution to this equation can be written

$$\Psi(t + \Delta t) = U(\Delta t) \Psi(t). \quad (11.150)$$

If H is time independent, the exact expression for the time-evolution operator is⁵⁰

$$U(\Delta t) = e^{-i H \Delta t / \hbar} \quad (11.151)$$

and for Δt small enough, the time-evolution operator can be approximated as

$$U(\Delta t) \approx 1 - i H \frac{\Delta t}{\hbar}. \quad (11.152)$$

While Equation 11.152 is the most obvious way to approximate U , a numerical scheme based on it is unstable, and it is preferable to use **Cayley's form** for the approximation:⁵¹

$$U(\Delta t) \approx \frac{1 - \frac{1}{2} i \frac{\Delta t}{\hbar} H}{1 + \frac{1}{2} i \frac{\Delta t}{\hbar} H}. \quad (11.153)$$

Combining Equations 11.153 and 11.150 we have

$$\left(1 + \frac{1}{2} i \frac{\Delta t}{\hbar} H\right) \Psi(t + \Delta t) = \left(1 - \frac{1}{2} i \frac{\Delta t}{\hbar} H\right) \Psi(t). \quad (11.154)$$

⁴⁹ This argument was introduced by B. Misra and E. C. G. Sudarshan, *J. Math. Phys.* **18**, 756 (1977). The essential result has been confirmed in the laboratory: W. M. Itano, D. J. Heinzen, J. J. Bollinger, and D. J. Wineland, *Phys. Rev. A* **41**, 2295 (1990). Unfortunately, the experiment is not as compelling a test of the collapse of the wave function as its designers hoped, for the observed effect can perhaps be accounted for in other ways—see L. E. Ballentine, *Found. Phys.* **20**, 1329 (1990); T. Petrosky, S. Tasaki, and I. Prigogine, *Phys. Lett. A* **151**, 109 (1990).

⁵⁰ If you choose Δt small enough, you can actually use this exact form. Routines such as Mathematica's **MatrixExp** can be used to find (numerically) the exponential of a matrix.

⁵¹ See A. Goldberg *et al.*, *Am. J. Phys.* **35**, 177 (1967) for further discussion of these approximations.

This has the form of a matrix equation $\mathbf{M} \mathbf{x} = \mathbf{b}$ which can be solved for the unknown $\mathbf{x} = \Psi(t + \Delta t)$. Because the matrix $\mathbf{M} = 1 + \frac{1}{2} i \frac{\Delta t}{\hbar} \mathbf{H}$ is **tri-diagonal**,⁵² efficient algorithms exist for doing so.⁵³

- (a) Show that the approximation in Equation 11.153 is accurate to second order. That is, show that Equations 11.151 and 11.153, expanded as power series in Δt , agree up through terms of order $(\Delta t)^2$. Verify that the matrix in Equation 11.153 is unitary.

As an example, consider a particle of mass m moving in one dimension in a simple harmonic oscillator potential. For the numerical part set $m = 1$, $\omega = 1$, and $\hbar = 1$ (this just defines the units of mass, time, and length).

- (b) Construct the Hamiltonian matrix $\mathbf{H}$ for $N + 1 = 100$ spatial grid points. Set the spatial boundaries where the dimensionless length is $\xi = \pm 10$ (far enough out that we can assume that the wave function vanishes there for low-energy states). By computer, find the lowest two eigenvalues of $\mathbf{H}$, and compare the exact values. Plot the corresponding eigenfunctions. Are they normalized? If not, normalize them before doing part (c).
- (c) Take $\Psi(0) = (\psi_0 + \psi_1) / \sqrt{2}$ (from part (b)) and use Equation 11.154 to evolve the wave function from time $t = 0$ to $t = 4\pi/\omega$. Create a movie (**Animate**, in Mathematica) showing $\text{Re}(\Psi(t))$, $\text{Im}(\Psi(t))$, and $|\Psi(t)|$, together with the exact result. *Hint:* You need to decide what to use for Δt . In terms of the number of time steps N_t , $N_t \Delta t = 4\pi/\omega$. In order for the approximation of the exponential to hold, we need to have $E \Delta t / \hbar \ll 1$. The energy of our state is of order $\hbar \omega$, and therefore $N_t \gg 4\pi$. So you will need at least (say) 100 time steps.



Problem 11.39 We can use the technique of Problem 11.38 to investigate time evolution when the Hamiltonian *does* depend on time, as long as we choose Δt small enough. Evaluating $\mathbf{H}$ at the midpoint of each time step we simply replace Equation 11.154 with⁵⁴

$$\left[1 + \frac{1}{2} i \frac{\Delta t}{\hbar} \mathbf{H} \left(t + \frac{\Delta t}{2} \right) \right] \Psi(t + \Delta t) = \left[1 - \frac{1}{2} i \frac{\Delta t}{\hbar} \mathbf{H} \left(t + \frac{\Delta t}{2} \right) \right] \Psi(t) . \quad (11.155)$$

Consider the driven harmonic oscillator of Problem 11.36 with

$$f(t) = A \sin(\Omega t) , \quad (11.156)$$

where A is a constant with the units of length and Ω is the driving frequency. In the following we will set $m = \omega = \hbar = A = 1$ and look at the effect of varying Ω . Use the same parameters for the spatial discretization as in Problem 11.38, but set

⁵² A tri-diagonal matrix has nonzero entries only along the diagonal and one space to the right or left of the diagonal.

⁵³ Use your computing environment's built-in linear equation solver; in Mathematica that would be $\mathbf{x} = \mathbf{LinearSolve}[\mathbf{M}, \mathbf{b}]$. To learn how it actually works, see A. Goldberg *et al.*, footnote 51.

⁵⁴ C. Lubich, in *Quantum Simulations of Complex Many-Body Systems: From Theory to Algorithms*, edited by J. Groendorst, D. Marx, and A. Muramatsu (John von Neumann Institute for Computing, Jülich, 2002), Vol. 10, p. 459. Available for download from the Neumann Institute for Computing (NIC) website.

$N_t = 1000$. For a particle that starts off in the ground state at $t = 0$, create a movie showing the numerical and exact solutions as well as the instantaneous ground state from $t = 0$ to $t = 2\pi/\Omega$ for

- (a)** $\Omega = \omega/5$. In line with the adiabatic theorem, you should see that the numerical solution is close (up to a phase) to the instantaneous ground state.
- (b)** $\Omega = 5\omega$. In line with what you've learned about sudden perturbations, you should see that the numerical solution is barely affected by the driving force.
- (c)** $\Omega = 6\omega/5$.